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DRAFT ENVIRONMENT AND SOCIAL SYSTEM ASSESSMENT (ESSA)

SOLAR ROOFTOPS FOR HOUSEHOLDS PROGRAM (P508489)

Prepared by

The World Bank

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EXECUTIVE SUMMARY

CHAPTER 1: PROGRAM DESCRIPTION

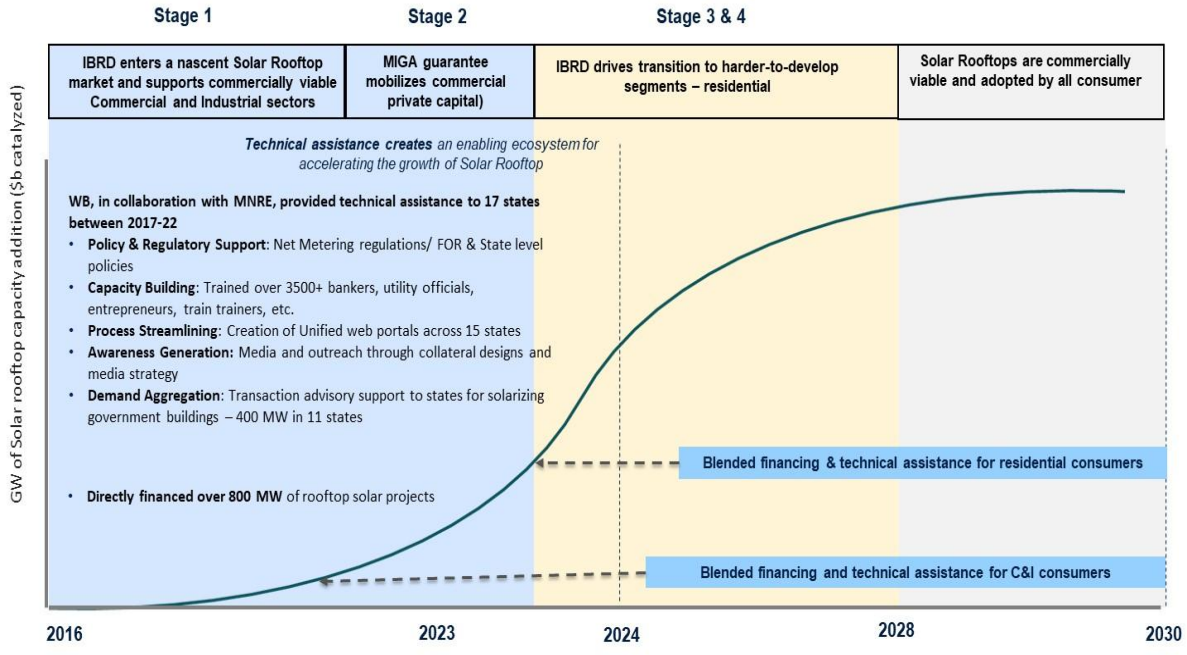
Context

1. The Government of India (GoI) has committed to ambitious energy and climate targets to contain future dependence on imported fossil fuels and GHG emissions growth, aiming to reach net zero by 2070. A cornerstone to achieving its net zero targets is sustainable growth in the power sector, with a particular emphasis on strengthening the role of renewable energy (RE) as it helps achieve the triple goals of energy security, affordability, and sustainability.

2. Achieving energy security, affordability, and sustainability would require (a) electrifying the end-user sectors (particularly the transport and building sectors), (b) greening the power sector with RE, and (c) bringing the hard-to-abate industrial sectors onto a low-carbon path with green hydrogen and carbon capture and storage (CCS). Electrifying the end-user sectors would further increase the electricity demand. Central Electricity Authority (CEA) estimates 500.6 GW of non-fossil fuel-based installed capacity, of which 292 GW is from solar PV and 100 GW by Wind by 2030, to meet the electricity demand of 2280 BU net of 35 BU generated by solar rooftops. The large-scale solar PV and wind sectors are well established with active private sector participation. The proposed Program supports the government in expanding residential solar rooftops, a sector that shows potential but faces challenges.

3. There is a strong rationale for the World Bank's support to the PM Surya Ghar: Muft Bijli Yojana (PMSG) given the Bank's extensive and longstanding engagement in the solar rooftop sector in India since 2016, which can be summarized in four distinctive stages:

Figure 1: S-curve for solar rooftop sector growth in India



(a) **Stage 1 (FY2016). World Bank partnered with the State Bank of India (SBI) to provide concessional financing for solar rooftop installations by C&I consumers, complemented by technical assistance (P155007):** The World Bank, together with the Clean Technology Fund (CTF), was the first risk-taker in the GRPV market, launching US$650 million Program-for-Results (PforR) with TA to the SBI. At the time, the GRPV market capacity was only 500 megawatts (MW), far below

the 40 GW target for 2022, and financing cost from banks was high, but it was brought down through the Program. This initiative directly mobilized US$265 million in private capital through borrower's equity. Within six years of the Bank's assistance, investments grew by over 900 percent - from US$500 million in 2016 to over US$5 billion in 2022.

(b) **Stage 2 (FY2023): Multilateral Investment Guarantee Agency (MIGA) refinanced part of the existing IBRD (World Bank) solar rooftop loan to SBI, mobilizing an additional US$200 million in private capital.** This refinancing underpinned the crucial role played by IBRD in initially de-risking the solar rooftops for C&I consumers and making them more attractive to private investors and facilitating subsequent investment with MIGA-backed guarantees.

(c) **Stage 3 (FY2024): World Bank extended support to GRPV in the Residential sector (P171750).** Building on the success of the previous solar rooftop Program and in response to the government's Phase II launch, the Bank approved an additional financing of US$165 million, expanding its financial support and TA through SBI to install 450 MW of solar rooftops across around 80,000 households. SBI provides discounted financing to consumers who otherwise would not have adopted GRPV systems. The initiative reduces financing costs market-wide and drives commercial investment in residential solar rooftops.

(d) **Stage 4 (FY2025): The World Bank is proposing a US$1 billion program to support the PM Surya Ghar: Muft Bijli Yojana**. The Bank aims to support the government's solar rooftop program by accelerating its implementation and preparation for its scale-up in the next phase through a new PforR operation. The operation would finance results to accelerate the deployment of solar rooftops for households and contribute towards achieving the Government's program's objectives by providing complementary TA[1] to MNRE, with special focus on incentivizing distribution companies to support the program and expand the benefits of the program to households with lower consumption, who are more difficult to incentivize because they pay highly subsidized electricity tariffs. From the perspective of Discoms and states, extending the benefits of solar rooftops to these households is highly beneficial as it would reduce the need for electricity subsidies. Other things being equal, the program, therefore, replaces a recurring tariff subsidy with a one-off investment subsidy, which is more financially sustainable.

Program Description

A. Program Development Objective(s) (PDO)

4. To increase the installed generation capacity of grid-connected rooftop solar photovoltaics by households and strengthen the capacity of relevant institutions to implement the PM Surya Ghar: Muft Bijli Yojana

B. Theory of Change and PDO Indicators

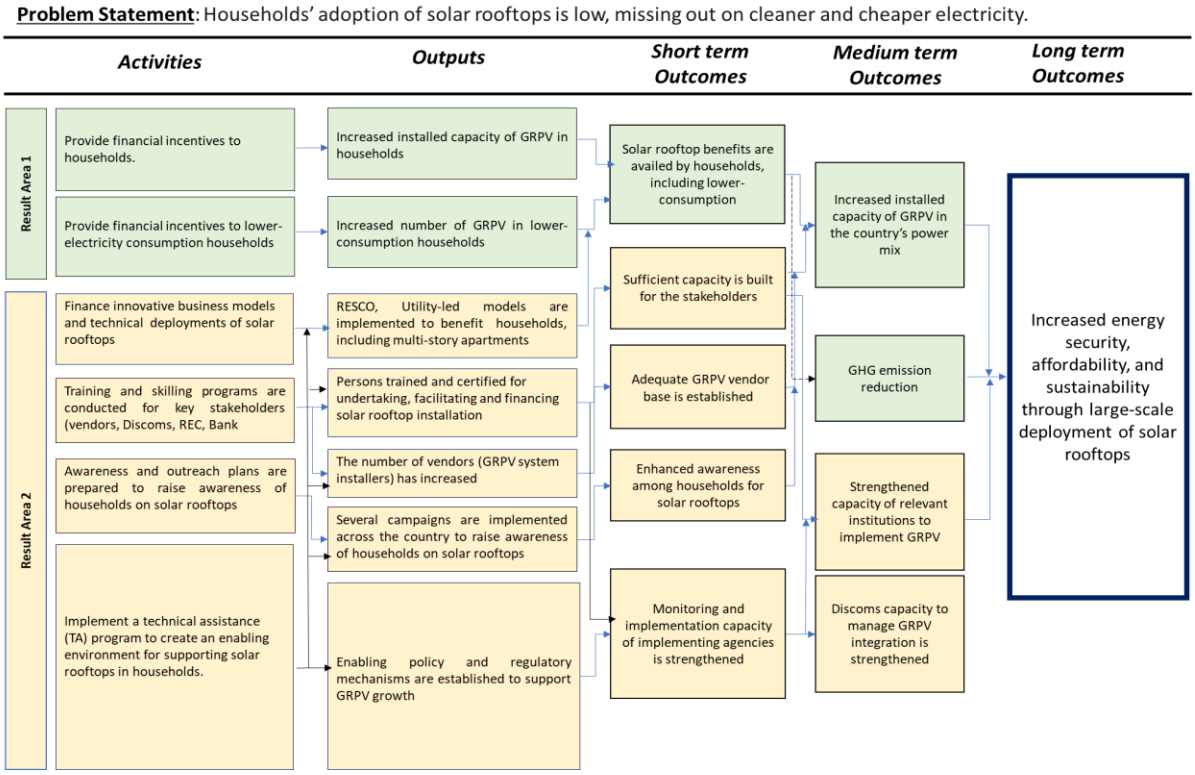
5. The Program's Theory of Change () highlights the activities needed to enhance the scaling-up of GRPV in the residential sector and reduce GHG emissions. The activities are structured as lending and technical assistance (TA). The financing will support results towards innovative business models and the GRPV systems installed by households. The TA will help raise consumer awareness of GRPV and its benefits, upskill officials of several agencies to perform their tasks better, facilitate low-income households' access to GRPV and the associated cost-savings on electricity bills, and establish mechanisms critical to the long-term sustainability of the PMSG program. These short-term outcomes will underpin the achievement of PDO outcomes and ultimately support India's energy transition by democratizing the use

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[1] The TA will be grant funded and recipient executed following PforR procedures. The team is engaging with several donors to source these funds.

of solar energy. The critical assumptions in this results chain are as follows: (a) continuity in the conducive policy and regulatory framework for GRPV installations in states; (b) there will be sufficient domestic manufacturing capacity for solar modules and cells along with sufficient installers to install the GRPV systems; and (c) banks and financial institutions willing to offer affordable loans to households for GRPV.

Figure 2: Theory of Change for the Proposed Program



6. The Program will track progress toward the PDO using the following proposed indicators:

a) Renewable energy capacity enabled with direct support (megawatt)

b) GHG emissions mitigated through household solar rooftop capacity addition (metric tons/year).

c) Persons trained and certified for solar rooftop installation through the PMSG Program (number).

C. PforR Program Scope

7. The government's program aims to contribute to India's NDCs by installing 30 GW of rooftop solar capacity through 10 million residential installations by March 31, 2027. The key objectives of the government's PMSG program are: a) achieving 10 million rooftop solar installations in the residential sector, resulting in b) self-generated electricity of up to 300 kilowatt-hours (kWh) per month per installation (assuming an average solar rooftop system capacity of 3kW), which is equivalent to c) a total of 1,000 TWh (billion kWh) of renewable electricity from the installed capacity, that will result in a reduction of 720 million tons of CO2e emissions during the 25-year lifetime of the rooftop solar projects. This is equivalent to saving over three-fourths of the country's emissions from coal-fired power plants for a year (2022-23).[2]

1.

[2] Coal-fired electricity output and emissions, Press Information Bureau, Government of India, available here: https://shorturl.at/kAe15

To achieve the scheme's objectives, the government is developing an enabling ecosystem for rooftop solar projects, including support to the regulatory framework, manufacturing facilities, supply chain, vendor network, and operation and maintenance facilities in the country. The program is also expected to boost the local economy and generate employment by encouraging investment in the local manufacturing of GRPV equipment and its installation by local vendors (usually small or medium enterprises), Discoms, and other renewable energy service companies (RESCOs), along with enhancing energy security and reducing household electricity bills.

8. **The PMSG program significantly boosts green jobs and spurs economic growth.** The International Renewable Energy Agency (IRENA) and the International Labor Organization (ILO) estimate that the renewable energy sector employs over a million people in India.[3] The government projects this program will create 1.7 million jobs in various sectors,[4] nearly half of the 3.5 million jobs needed to meet the 500 GW non-fossil electricity generation goal by 2030.[5] Small-scale solar energy, particularly solar PV and distributed renewable energy, generates more jobs in India. With an average age of 27, India's economic growth and employment are critical issues now and in the future. The PMSG program is crucial for shaping a greener future, supporting energy independence, environmental sustainability, and creating employment opportunities nationwide.

9. **The PMSG program introduces several measures to facilitate the adoption of GRPV by residential consumers.** It provides capital subsidies for solar rooftop system capacity up to 3 kW capped at about US$926 (INR 78,000) for a 3kW system (60 percent of the benchmark capital cost for up to 2 kW system and 40 percent for system capacity between 2 and 3 kW; for systems higher than 3kW, the subsidy would be limited to first 3kW, significantly lowering upfront investment costs. The program also includes provisions for collateral-free loans and features a streamlined online portal, thus further incentivizing solar rooftops and providing a seamless consumer experience in installing solar rooftops. MNRE is leading and supervising the implementation of the program, while REC Ltd. (formerly Rural Electrification Company)[6] has been nominated as the national-level program implementation agency (NPIA), REC, in charge of transferring payments to program beneficiaries after verifying that their obligations have been fulfilled. Table 3 summarizes the government's PMSG program components and budgeted costs.

10. The proposed PforR Program ('P,' the Program) will support components 1, 5, 7, 8, and 9 under the government's program (small 'p') in the Program Area. The Program is designed around two mutually reinforcing results areas (RA) to build a solid foundation to achieve the long-term goal of increased energy security, affordability, and sustainability through large-scale deployment of solar rooftops. Activities under both RAs will be carried out concurrently, with disbursements tied to achieving Disbursement Linked Indicators (DLI) throughout the Program cycle. The Bank's PforR supports the MNRE and REC Ltd in better implementing the government program by specifically incentivizing measures, including a

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[3] IRENA and ILO (2024), Renewable energy and jobs: Annual review 2024, International Renewable Energy Agency,

Abu Dhabi, and International Labor Organization, Geneva.

[4] Cabinet approves PM-Surya Ghar: Muft Bijli Yojana for installing rooftop solar in One Crore households, Press Information Bureau, Government of India, accessed here: https://shorturl.at/7twzL

[5] Akanksha Tyagi et.al, 2022. India's Expanding Clean Energy Workforce. Council on Energy, Environment and Water, Natural Resources Defense Council, and Skill Council for Green Jobs.

[6] REC is a 'Maharatna' company under the Ministry of Power's (GoI) administrative control. It is registered with the Reserve Bank of India (RBI) as a Non-Banking Finance Company (NBFC), Public Financial Institution, and Infrastructure Financing Company. (Maharatna translates to a great jewel; this status is provided to public-sector companies with consistent overachievement of high financial performance thresholds.

comprehensive technical assistance program, that addresses the PMSG program barriers (as mentioned in paragraph 12). Government program components 2, 3, 4, and 6, have been excluded from the Bank's Program boundary to limit the administrative efforts of managing and monitoring a nationwide program. The following activities are excluded from the PforR Program: investments that may present high environmental and social (E&S) risks, subject to confirmation before appraisal, and with contract values exceeding the PforR threshold, to be determined before appraisal.

11. **RA1: Expand solar rooftop generation in households.** The PMSG program's potential residential consumer beneficiaries are households with higher- and middle-electricity consumption. By starting at these consumer levels (with the GoI program targeting 10 million HHs out of more than 200 million technically feasible HHs in the country),[7] the Bank's Program is incentivizing a sub-sector, including the solar PV supply chain, as it unfolds, allowing for the benefits to seep in, like rooftop solar system cost-reduction, paving the way for other consumers (particularly the low electricity consumption HHs) to benefit from solar rooftops likely in future years (beyond the current program). Currently, [90] percent of households who have installed solar rooftops under the program have an electricity-connected load exceeding 3 kW, representing a much smaller share of the households as most have < 3 kW connected load.[8] The unique characteristics of such consumers across states in India make it difficult to target these consumers. First, at least eight states provide between 100 and 300 electricity units (kWh) free per month, creating no incentive for those consumers to adopt solar rooftops. Second, even if the states are not providing free electricity, these consumers' average electricity tariff is very low (usually below INR 4/unit or US0.04/kWh) to make electricity affordable for them, rendering the solar rooftop commercially unviable even with 60 percent subsidies under the government program. Third, many of these households either don't own roofs (in the case of multi-storied apartments) or have roofs unsuitable for solar rooftop installations. Fourth and most importantly, vendors/ solar rooftop system installers are not looking to tap these consumers (<3kW), given the weak economies of scale based on our feedback from consultations conducted by the team. However, to align with the MNRE objective of encouraging all households to join the program, and considering that Discoms and states benefit from improved financial performance or reduced subsidy burden if low-electricity consumers switch to solar rooftops, the Bank's Program will incentivize households that consume less electricity (defined as less than 300 kWh per month, corresponding to an electricity-connected load of up to 3 kilowatts (kW)) through a separate DLI#2. This DLI is linked to DLI#3, which promotes innovative business models for reaching lower-electricity consumers, and to DLI#6, where technical assistance will help design and implement these models. DLI#4 contributes to the second part of the PDO: "strengthen the capacity of relevant institutions to implement the PM Surya Ghar: Muft Bijli Yojana," the relevant includes Discoms, vendors, REC Ltd., and banks.

12. RA2: Support market development, strengthen institutional capacity, create an enabling environment, and foster sustainability of the PMSG program. Under this RA, the focus will be around:

(a) Market development through focus on (i) developing and implementing market aggregation models and innovative business models such as utility-led approaches and RESCO models mainly targeting low-electricity consumption households and states and geographies struggling to adopt GRPV.

(b) Strengthening the capacity of different stakeholders for rooftop installation through the PMSG program in the Program Area, including building a capable local skilled workforce (technicians/ electricians/ installers/ engineers/ supervisors/ users/ vendors, etc.). In addition, capacity building of other relevant stakeholders is crucial, including DISCOM officials, banking/financial institution officials, central and state government officials (for effective convergence of efforts), state renewable energy development

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[7] Total number of technically feasible HHs are computed by assuming a GRPV system size of 3kW and 637 GW of GRPV potential estimated by CEEW in its study Mapping India's Residential Rooftop Solar Potential, November 2023.

[8] All India Electricity Statistics, General Review 2024, Central Electricity Authority, GoI, May 2024 (Refer to Table 10 on pp.177)

agencies, etc. To support implementation, Skilling and Capacity Building Action Plans will be implemented and monitored at the national and state levels.

(c) Promoting Awareness and Outreach. Installation of 10 million rooftops in the residential sector would require creating awareness and disseminating information regarding the benefits available under the PMSG program. Under this DLI, the focus will be to utilize various channels for promotion to make awareness about the installation of GRPV, their registration and application process, knowledge dissemination on good practices, and information dissemination to the public regarding various components of the scheme. The NPIA will implement and monitor this component for the scheme at the national level.

(d) The TA program will primarily support MNRE through advisory support and knowledge exchange visits to effectively implement different components of the PMSG program by strengthening or reinforcing human resources at the NPIA. In addition, the TA will support the various emerging and dynamic needs of MNRE, which would require analytical and knowledge support. This could encompass the following thematic areas and the indicative activities but is not limited to:

- *Programmatic support to selected states / DISCOMs*: Implementation (knowledge) support to states, including adopting the new or existing model regulations or business models.

- *New approaches for better integration and management of solar rooftops by Discoms*: (i) Technical solutions (develop IT tools and instruments) for demand forecasting under a high share of solar rooftops, managing load profiles, etc. (ii) Prediction tools for establishing mechanisms for better DRE management within Discoms.

- *Skilling and upskilling programs for crucial stakeholders*: (i) Skilling—Content generation (*improving* the quality and delivery of training modules using digital learning aids, lectures, Virtual Reality training for solar rooftop installers, etc.). (ii) Support the development of existing institutions into dedicated solar rooftop training academies and create a Centre of excellence.

- *Raise consumer awareness*: (i) Update information, education, and communication (IEC) materials to raise consumer awareness. (ii) Support better design of *consumer outreach programs.*

- *Analysis of domestic solar PV module supply chain:* provide price and availability insights to vendors and recommend policy measures to the ministry.

- *IT Solutions for digitalizing public sector banks' solar rooftop loan processing:* the aim will be to understand the current IT system deployed at these banks and provide external IT support to help them digitalize the process.

- *A mid-term review of the PMSG program implementation to identify good practices, lessons learned, and any gaps that need to be addressed, which can provide input for* the design of the next phase/ scale of the program.

Table 1: The government solar rooftop program and the Bank's Program boundary

	Government program	**Program supported by the PforR (PforR Program)**	**Reasons for non-alignment**
Objective	To increase the share of solar rooftop capacity and empower residential households to generate electricity to meet their demand.	To increase the installed capacity of Grid-connected Rooftop Solar Photovoltaics (GRPV) in households and strengthen the capacity of relevant institutions to	Broadly aligned, but the Bank Program emphasizes increasing the proportion of households with low electricity

	Government program	Program supported by the PforR (PforR Program)	Reasons for non-alignment
		implement the PMSG program.	consumption who install solar rooftops through the program, which is not explicitly targeted under the GoI program.
Duration	2024- 2027	2025-2028	An additional year under the Bank Program is needed to ensure full disbursement of funds and the monitoring of results.
Geographic coverage	Nationwide	Program Area will be defined in the Program Operations Manual (POM).[9]	-
Results areas	1) Supporting the increased capacity of solar rooftops in households. 2) Discoms are incentivized to facilitate solar rooftops in their network. 3) Model solar villages are created in each district. 4) Urban local bodies are incentivized to support the growth of solar rooftops in their jurisdiction. 5) RESCO and Utility-led models are supported to benefit consumers without adequate roofs or roof rights. 6) Foster innovation in technology, business models, and financial instruments 7) Strengthening institutional capacities 8) Increasing consumer awareness of solar rooftops	1) Expanding the adoption of solar rooftops by households, including those with lower electricity consumption. 2) Support market development, strengthen institutional capacity, create an enabling environment, and foster sustainability of the GRPV program	The PforR RAs are consistent with the government program but regrouped into two pillars. Excluded activities are summarized in paragraph 12 above.
Overall Financing	US$8.9 billion (excluding any state government complementary financial assistance)	US$1 billion	The PforR Program finances 11 percent of the government program.

1.

[9] Program Area will be defined in accordance with Bank policies.

13. **Program Beneficiaries.** The direct Program beneficiaries are the households that will get lower-cost electricity from solar rooftop systems. Discoms will benefit from lower electricity losses and less needed subsidies to relevant households, which implies better financial health for the Discoms. This, in turn, will support more private-sector investment in the sector as the off-taker risks are mitigated through improved financial health of Discoms. An estimated [300,000] technicians will be trained through the program, thus enabling them to get jobs with vendors (system installers) or become entrepreneurs by setting up their solar rooftop installation companies. The entire community will benefit from the shift from essentially coal-fired thermal power to solar generation, resulting in lower emissions of pollutants and better air quality. Reduction in GHG emissions will also benefit the global community through climate change mitigation.

14. **Program Financing.** The total program financing of US$12.71 billion comprises public expenditures of US$7.91 billion and US$3.8 billion of commercial funds. The public expenditures come from the government of India's budgetary allocation for the PMSG program, of which US$1 billion is proposed from the IBRD and CTF financing combined. The commercial funds are expected to be mobilized by commercial banks in India that provide loans to residential consumers or RESCO players for installing solar rooftop systems. The computation assumes a typical system size is 3kW, costing US$1,214 net of government incentives. On average, based on the current information with MNRE, 40 percent of the households are expected to opt for commercial loans for 80 percent of the system costs net of government incentives.[10]

Table 2: Program Financing

Source	**Amount (US$, millions)**	**Percent of Total[11]**
Household funding – from commercial sources	3,800	30
Government of India[12]	7,910	62
IBRD (proposed)	940	7.5
Clean Technology Fund Loan (proposed)	60	0.5
Total Program Financing	12,710	**100.00**

15. **Program Expenditure Framework (PEF).** The Program expenditure framework will include a slice of expenditures under component 1 (first 5 million residential customers starting from xx date) and fully include component 5 under the government program as described in Table 3. In the case of the expenditure under components 7, 8, and 9, only the activities being implemented by NPIA (REC) will be covered starting from xxx date. The expenditure under component 9 will cover the Program establishment and operating costs of REC, including PIU expenses, engagement of crucial staff required to operate the

1.

[10] Formula: PCM=$1,214 (prices for a 3kW solar rooftop system net of subsidy) X 80% (of the capital cost net of subsidy) X 40 million (number of households expected to avail commercial loan for financing their rooftop solar system).

[11] Rounded off to single decimal values

[12] GoI is also in discussion with other MDBs to raise part of the financing for the government program, including the Asian Development Bank (ADB) for a US$1 billion policy lending operation.

Program, various consultancy services, audit fees, legal service fees, and independent verification agent (IVA) charges.

Table 3: Components of the government's PMSG program

C. No.	Component Name	Scheme Allocation		PEF (estimated)		Remarks
		INR (crores)	US$ million	INR (crores)	US$ million	
1	Central Financial Assistance (CFA) to Residential Consumers	65,700	7,800	32,850	3,900	For the first 5 million beneficiaries (release of funds to the beneficiaries starting from xx date)
2.	Incentives for distribution companies	4,950	590	0	0	
3.	Model Solar Villages in each district	800	95	0	0	
4.	Incentives for Local Bodies	1,000	119	0	0	
5.	Payment Security Mechanism	100	12	100	12	This will be operationalized once implementation arrangements are detailed in POM.
6.	Innovative Projects	500	60	0	0	
7.	Capacity Building (1 percent of CFA)	657	78	328.5	39	Expenditure on activities implemented by REC at the national level starting from xxx date.
8.	Awareness & Outreach (1 percent of CFA)	657	78	328.5	39	
9.	Service Charge (1 percent of CFA)	657	78	65.7	7.8	Operating expenditure to be implemented by REC at the national level starting from xxx date, including those on IVA and auditor's costs.

	Total	**75,021**	**8,910**	**33,672.7**	**3,997.8**	

D. Disbursement Linked Indicators

Table 4: Disbursement Linked Indicators

Purpose of DLI	**DLI**	**Financing allocated from IBRD, CTF, and ESMAP (US$, Millions)**
RA1: Expand solar rooftop generation in households		**695**
Expanding solar rooftop adoption in households	DLI 1: Grid-connected solar rooftop capacity installed at households in the Program Area (Megawatt)	IBRD: 340 CTF: 55
Expanding solar rooftop adoption in the Lower Consumption Households	DLI 2: Grid-connected solar rooftop systems installed in Lower Consumption Households in the Program Area (number)[13]	IBRD: 300
RA2: Support market development, strengthen institutional capacity, create an enabling environment, and foster sustainability of the PMSG program.		**305**
Develop and promote new business models for solar rooftops, including utility-led and RESCO models	DLI 3: The Mission Directorate of PMSG has approved new business models, including utility-led aggregation and RESCO models, under the PMSG program in several states in the Program Area (number).	IBRD: 150
Capacity building and skilling of solar rooftop installers	DLI 4: Persons trained and certified for solar rooftop installation through the PMSG program in the Program Area (number).	IBRD: 75
Promote awareness and outreach among households	DLI 5: The Mission Directorate of PMSG has approved Awareness and outreach plans (number)	IBRD: 75
Technical assistance	DLI 6: The Mission Directorate of PMSG has adopted and implemented a technical assistance action plan for key stakeholders for implementing the PMSG program.	CTF: 5[a]
		Total: 1000

Note a. The task team has contacted FFI to explore accessing grant funds to implement TA.

E. Role of Partners

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[13] Lower Consumption Households are defined as <3kW contracted/ connected load.

Table 5: Role of Partners

Name of Partner	Nature of Involvement /Description
Asian Development Bank (ADB)	GoI is also discussing with ADB to raise US$1 billion of financing for the government program. Based on the initial discussions with DEA/ADB, ADB is preparing a policy lending operation.
International and national think tanks and donors	Several of the international and domestic think tanks and donors are partnering with MNRE to provide technical assistance and analytical support in their areas of expertise and in specific geographies based on available resources. These include (non-exhaustive), GIZ, Shakti Sustainable Energy Foundation, Bloomberg Philanthropies, Council on Energy, Environment and Water (CEEW), Center for Study of Science, Technology and Policy (CSTEP), India Smart Grid Forum (ISFG), Natural Resources Defense Council (NRDC), National Solar Energy Federation of India (NSEFI), Rocky Mountain Institute (RMI), and Vasudha Foundation.

Need for Environmental and Social Systems Assessment (ESSA)

16. In keeping with the World Bank's guidelines for PforR project preparation, a comprehensive assessment of environmental and social systems (ESSA) has been undertaken. The objective of the ESSA is to ensure consistency with the core principles and directives on Program-for-Results Financing in order to effectively manage program risks and promote sustainable development.

17. The purpose of the Environmental and Social Systems Assessment (ESSA) is to: (i) review the environmental and social management rules and procedures and institutional responsibilities that are being used by the Program; (ii) assess the implementing agency (MNRE's) institutional capacity and performance to date to manage potential adverse environmental and social issues under the Program; and (iii) recommend specific actions for improving the capacity of MNRE in regard to effective management of environmental, health, safety and social issues during implementation.

Institutional and Implementation Arrangements

18. ***Ministry of New and Renewable Energy (MNRE)*** is responsible for formulating, implementing, and monitoring the PMSG program. The Mission Directorate in MNRE will implement the scheme. The Directorate will be headed by a Mission Director, not below the rank of Joint Secretary to the Government of India.

19. ***National Program Implementation Agency (NPIA):*** REC Limited has been appointed the national implementation agency; however, it functions under the direction of the Mission Directorate. REC operates the national portal, including vendor registration at the national level, and undertakes government program-related activities. REC Limited is also the Central Nodal Agency (CNA) for the PMSG program (CNAs for any government scheme are responsible for managing the funds off or on the balance sheet; as a CNA, it requisitions, disburses, and ensures timely and transparent funds flows to the beneficiaries and other implementing agencies.

20. Group of Ministers: A Group of Ministers will be constituted for guidance and coordination with states.

21. Steering Committee: A Steering Committee under the chairmanship of the Cabinet Secretary and including Secretaries of the Ministry of New and Renewable Energy, the Department of Financial Services, the Ministry of Power, the Ministry of Housing and Urban Affairs, and the Ministry of Rural Development will monitor the implementation of the scheme.

22. ***State Implementing Agencies (SIAs):*** The Distribution Utility (DISCOMs or Power/Energy Departments, as the case may be) shall be the SIA at the State/Union Territory level and shall ensure adherence to timelines specified for the provisioning of services for households and shall coordinate with REC on matters of the performance of vendors registered on the National Portal, grievance redressal, program monitoring at State/UT level. The Program Operations Manual (POM) will specify the respective roles of each MNRE and REC and the Program Area of implementation.

23. State Level Coordination Committee: There will be a State Level Coordination Committee headed by the Chief Secretary/ Advisor to the Lieutenant Governor or Administrator, and comprising Secretaries of key departments to coordinate and monitor the scheme's implementation in the State/Union Territory.

24. District Level Committee: There will be a District Level Committee headed by the District Magistrate or District Collector and comprising other members to monitor the scheme's implementation at the district level.[14]

25. ***Eligible Consumer:*** Any interested consumer with a valid Consumer Account Number (or its equivalent consumer ID) for a Distribution Utility (Distribution Companies, or in some cases Power/Energy Department of the state wherever applicable) shall apply on the National Portal.

26. ***PV Developers (Vendors), RESCOs and Utility Led Aggregators (ULAs):*** The installation of solar panels under the PMSG will be done only by registered vendors/RESCOs/ULAs engaged by the consumer through the National Portal.

27. ***Monitoring:*** MNRE is setting up a multi-tier mechanism for monitoring the implementation of the government program. A Group of Ministers will be constituted for guidance and coordination with states. A Steering Committee under the chairmanship of the Cabinet Secretary and including Secretaries of the Ministry of New and Renewable Energy, the Department of Financial Services, the Ministry of Power, the Ministry of Housing and Urban Affairs, and the Ministry of Rural Development will monitor the implementation of the scheme. There will be a State Level Coordination Committee headed by the Chief Secretary, Advisor to the Lieutenant Governor or Administrator, and comprising Secretaries of key departments to coordinate and monitor the PMSG program implementation in the State/Union Territory. There will be a District Level Committee headed by the District Magistrate or District Collector and comprising other members to monitor the implementation of the PMSG program at the district level.[15]

ESSA Approach

28. The ESSA is a World Bank document requirement for PforR operations. It is prepared by the Bank through a combination of reviews of existing program materials and available technical literature. ESSA methodology requires consultations with MNRE, key stakeholders and field visits to understand implementation nuances and uptake of the government program. However, limited interaction of stakeholders including virtual consultation with vendors only and field visits in National Capital Region, both facilitated by Bank, have been made possible till March 31, 2025. The findings, conclusions, and opinions expressed in the ESSA document are those of the Bank. Recommendations contained in the analysis shall be discussed and finalized with the MNRE officials.

29. This ESSA is prepared primarily based on a) review of existing literature and documents available with MNRE (provided by them or on the program website; the program guidelines, etc.) and other

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[14] Operational guidelines for implementation of PM-Surya Ghar: Muft Bijli Yojana for the component “Central Financial Assistance to Residential Consumers,” MNRE, GoI, June 7, 2024.

[15] Operational guidelines for implementation of PM-Surya Ghar: Muft Bijli Yojana for the component “Central Financial Assistance to Residential Consumers,” MNRE, GoI, June 7, 2024.

secondary literature publicly available and b) limited interactions with stakeholders;1) physically one residential household at Delhi one Resident Welfare Association at Delhi both serviced by a single vendor and 2) virtual meeting with 22 vendors (7 physically and about 15 virtually participating) during a virtual meeting organized by The World Bank. Therefore, the ESSA draws upon ongoing assessments, secondary literature, meetings with one household and one residential welfare association and one virtual meeting with about 22 vendors. The ESSA largely focused on social and environmental systems and institutional capacity to manage the identified risks and provides recommendations for strengthening these systems and capacities. This program is unique because it is spread across the India and involves several stakeholders. The methodology included a desk review followed by a mapping of stakeholders (both direct and indirect beneficiaries).

30. Desk review covered analysis of existing policy, operational procedures, institutional capacity, and implementation effectiveness relevant to the activities under the Program. Range of environmental and social issues like occupational health and safety workers and public health and safety, labour welfare, social inclusion, gender, and citizen engagement were examined. Existing documents, documents and reports received from MNRE program website, data, etc. were reviewed.

31. The methodology for preparation of the ESSA involved the following:

(i) a review of the government program material and technical literature to assess the program's potential to address environment and social issues including its review against the six core principles outlined in the Bank Policy or Program for Results financing;

(ii) a desk review of the laws, regulations, requirements, and guidelines on the Environment Health Safety (EHS) and social management;

(iii) interactions with large vendors who have been involved in the business of setting up rooftop Solar PV systems;

(iv) field visits to a group housing society (GHS) and individual household in National Capital Region;

(v) based on any identified gaps, a proposed Program Action Plan (PAP) has been devised;

32. **Disclosure:** The stakeholder consultations will be carried out to update the draft ESSA which will be publicly disclosed by MNRE and Bank for dissemination of ESSA through a multi-stakeholder consultation workshop. The final ESSA report will be disclosed on the WB's external website and the MNRE's website after incorporating the feedback of the workshop.

CHAPTER 2: STAKEHOLDER CONSULTATIONS

1. The following are the likely stakeholders associated with the project:

   a) MNRE (Lead implementing agency responsible for overall program execution, policy formulation, and regulatory oversight)
   b) RECL (Implements the program at the national level, coordinates with state agencies, disburses funds, and monitors implementation)
   c) RESCOs
   d) DISCOMs
   e) State Nodal Agencies (Assist in policy implementation at the state level and support consumer awareness and training)
   f) Urban Local Bodies (Facilitate rooftop solar adoption at the local level, engage communities, and ensure implementation in urban and rural areas).
   g) Commercial Banks and Financial Institutions - Provide loans and financing mechanisms for households adopting solar rooftops.
   h) Private Sector (Renewable Energy Service Companies - RESCOs, Solar Installers, and Manufacturers) - Install and maintain solar rooftop systems, support market development, and contribute to local manufacturing of solar components.
   i) Empaneled Agencies (Vendors) (Sample)
   j) Low Income Household- - Target beneficiaries to achieve DLI
   k) Residential Welfare Associations of High Rise Buildings
   l) RWAs of LIG Housing Societies
   m) BPL Communities
   n) Tribal Communities (including PVTG Areas)
   o) Vulnerable Groups (Low Participation)
   p) Present Beneficiaries/ Households (End Users/Beneficiaries) -- Participate in the program by installing rooftop solar systems and benefiting from lower electricity costs.

2. Limited stakeholder consultations have been carried out so far. Field visits to a GHS and an independent household were undertaken in National Capital Region. Consultations with vendors through a meeting dedicated to discussing environmental and social management issues was held. Both interactions were organized by the World Bank through the ongoing program with State Bank of India (Grid Connected Rooftop Soal Program-P171750). Only large vendors who work in multiple states participated in the consultations.

3. **Consultations for ESSA:** limited interactions with stakeholders;1) physically one residential household at Delhi one Resident Welfare Association at Delhi both serviced by a single vendor and 2) virtual meeting with 20 vendors (7 physically and others virtually participating) during a virtual meeting organized by The World Bank. However, more consultations are foreseen as the preparation progresses. Also, checklists to collect information on environment and social aspects associated with the stakeholder organizations/ departments/ agencies were shared and the responses are yet to be received. The summary of these consultations are given below:

4. **Field Visits in National Capital Region**

   a. **Individual User in Lajpat Nagar-II, New Delhi:** The user has a load of 7 KW and installed 3.24 KW solar rooftop PV by hiring the services of a registered vendor from the MNRE portal.

i. Implementation Process: The consumer confirmed that all interactions with DISCOM staff and procurement-related activities were carried out by the vendor, including due diligence and safety inspections of the rooftop site. However, no official records of the due diligence conducted by the vendor's staff have been maintained, nor has any report been submitted to MNRE.

ii. Fire Safety: Provision of fire extinguishers are implemented by the vendor as best practices but are not mandated under the contract agreement.

iii. Panel cleaning: Similarly, the service of cleaning of panels is being provided by the vendor but is not part of his contract.

b. Jagaran CGHS, Dwarka, New Delhi: The installed solar rooftop system has a DC capacity of 60 kW and an AC capacity of 50 kW, with solar generation recorded monthly by DISCOM. The system comprises bifacial 575W Topcon panels.

i. Implementation Process: The vendor conducted all due diligence, including conducting a preliminary physical assessment of the rooftop and preparing structural drawings. The structural designs proposed by the vendor were reviewed and approved by the RWA to ensure the safety of the installation.

ii. Fire Safety: Provision of fire extinguishers are implemented by the vendor as best practices but are not mandated under the contract agreement.

iii. Panel cleaning: No provision of dedicated water pipeline for cleaning is existing and water from fire tank hose is being used.

5. **Meeting with Vendors:** A hybrid consultation was organized with the vendors by the World Bank in its office. Larger vendors engaged in implementing the government program in multiple states across the country participated in the meeting. They primarily work through channel partners, including those from outside the solar PV sector, who are interested in supporting the program despite lacking prior experience. Currently, there are about 12,000 solar channel partners; out of these 90% are new entrant with potential for becoming partners.

6. The detailed feedback from the vendors is included in Annexure 3. Concerns shared by vendors on environmental and social management include:

- Site Selection: Lack of clear guidelines on environmental considerations such as roof strength, shadow-free area, and ecological impact.
- OHS: Adherence to OHS protocols by workers is not mandatory and is vendor specific. Some of the vendors have manuals and standard operating procedures for health and safety, but it is not a general practice.
- Roofing Materials and Drainage: Inadequate assessment of roofing material, compatibility and proper drainage systems are leading to potential structural safety issues. No assessment is done for presence of carcinogenic material, like asbestos, in the roof where installation will be done.
- Provision of dedicated water source for panel cleaning is required for effective maintenance of the panels.
- Maintenance and Monitoring: No clear framework for periodic inspections to ensure ongoing compliance with safety standards. The maintenance tasks is undertaken by a different team within the vendor agency, primarily responsible for cleaning of the panels on a periodic basis. All other safety aspects or repairs are undertaken on a case-to-case basis.
- Waste Management: There are no guidelines for disposal and recycling of installation-related waste (e.g., packaging materials, damaged panels). As per the E-waste Management Rules, 2022 amendment, the manufacturer is required to take back and store the PV panels. The vendor takes responsibility in

some of the cases to liaison with the manufacture to return the panels. This arrangement is feasible for the vendor for the O&M duration of 5 years only. Additionally, there is clear policy on the responsible disposal or recycling of end-of-life solar panels, batteries, and inverters.

- Training on EHSS: The capacity building component of the government program lacks structured program for training workers, vendors, and DISCOM officials on environmental safeguards.
- Public Awareness: There are no adequate mechanisms and efforts to educate end users on the environmental benefits and risks associated with solar rooftop installations.

7. Certain large vendors, such as TATA, Luminous, and Redington, have established Standard Operating Procedures (SOPs) that incorporate fundamental environmental safeguards, including worker safety, responsible waste management, and adherence to site selection guidelines. These practices help mitigate potential environmental and occupational hazards during solar rooftop installations.

8. Tata Power leverage their own application to validate roof structures and ensure safety using AI technology. The system provides an instant sanction or rejection based on quality standards, requiring any negative markings to be rectified until a 100% score is achieved. A warranty certificate is issued only upon meeting these standards, making it an integral part of the incentive structure for channel partners. As a result of these initiatives, they have received minimal complaints in the residential sector. O&M activities are being carried out separate teams

9. Social Issues: The following social issues were expressed by the participants and/or noted during the virtual meeting:

a) There are several complaints about the solar portal; some of the key complaints are a) the portal is regularly down, b) subsidy payments are over delayed, c) the menu to download customer data are disabled, d) there are no replies from MNRE for the complaints, e) no feedback is collected from customers, f) registrations take a lot of time and consumers are unable to register, g) for filing any grivances the vendors and customers are unable to reach MNRE, etc,. Since some customers have not received their subsidies, they are blaming the vendors for this and sending legal notices to the vendors and also making complaints to the Chief Ministers, Prime Minister, MLAs, MPs, etc. and so many other people.

b) Obtaining the required certificates after new guidelines are issued is posing a problem for the customers and also for the vendors. Customer Support is required at MNRE level. Presently no replies are given to any complaints from MNRE.

c) Only UP, Maharashtra Gujarat, etc. states are doing very well in achieving targets, whereas the East and North East states are not performing well. This need to be taken care of by providing awareness to the beneficiaries and removing all the obstacles blocking the progress.

d) Presently, the project is around 80% in urban areas and around 20% in rural areas; this need to be changed and more rural projects need to be taken up to meet the target.

e) The Vendors say for the safety of the workers, they provide them with PPE kits, all the required tools, instruments for measuring us voltage and resistance, etc. are provided.

f) The vendors say that most of the workers are either ITI certificate holders or diploma holders or had some training in the Solar installation. The workers are also given some induction training. Some of the vendors are CII certified.

g) The Vendoers say they have registraion. They say there are some small scale vendors who who may not have such kind of registrations are certifications presently.

h) Some of the DISCOM's don't even inspect this sites for monitoring/ inspection/ commissioning/ certification.

i) The Vendors say that the empanelled vendors are having labour licenses as per the state level norms. The labour insurance is something that some vendors have and some vendors or yet to take depending upon the kind of contracts they have with their workers.

j) Registration of migrant labor is not taken up by the vendors.

k) The vendors say they are paying the workers for accommodation during outstation duties when they go on site.

l) Safety tests such as Vetigo Test for workers working at heights are yet to be introduced. As per the vendors, in general there is some training or induction training given to the workers. They say so far no accidents are reported.

m) The vendors say no female workers are working on site, but they work as office staff at the offices. They say GBV training is given by some of the vendors to their workers. But they do not have any GBV training material of their own. They have to start formation of ICs.

n) The vendors inform that an informal internal GRM is in place for workers.

o) The vendors say they will be sharing the checklist that DISCOMs use for approving the project.

p) Some of the vendors say they have manuals and standard operating procedures for health and safety, but some of them do not have. They will share these with the Bank.

CHAPTER 3: ENVIRONMENTAL AND SOCIAL EFFECTS

Introduction

1. A Program Action Plan will address EHS concerns associated with GRPV installations, ensuring that risks are minimized, and the program operates within acceptable environmental and social parameters.

Environmental Effects

2. The overall potential environmental effects of the Program will be positive. The Program supports reduced reliance on fossil fuels thereby reducing greenhouse gas emissions, improving air quality and aligning with India's commitments under the Paris Agreement for climate mitigation. The investments to be financed under the Program – elevated ground and roof mounted GRPV installations - are not expected to have significant environmental risks and impacts and can be mitigated through effective management. The government program estimates that the scheme will generate 1000 BUs of electricity while reducing CO2 emissions by 720 million tons over the 25-year lifetime of these PV systems. Over the 25-year lifetime of these rooftop systems, it is estimated, making a substantial positive impact on the environment[16].

3. The environmental impacts will vary governed by factors such as site selection, scale of installation, and management practices. The proposed investments can be considered as minor modifications to existing facilities. The environmental effects are small or modest in intensity, of limited duration and extent, mostly completely reversible, and can be readily mitigated through cost-effective measures. Potential investments will not encroach or degrade sensitive habitats, nor be in sensitive areas of bio-diversity value, nor located in areas protected for physical cultural resources.

4. The environmental health and safety (EHS) concerns associated with GRPV specific requirements of the Program are as follows:

(i) Site Selection

Level of Concern: High

Potential investments encroaching or degrading sensitive habitats, sensitive areas of biodiversity value, or located in areas protected as physical cultural resources.

Mitigation Measures: Such sites will have to be identified by consumers with the help of Urban Local Bodies (ULBs) and competent authorities during evaluation of proposals. Sites falling in such areas will not be implemented through the Program and will be included in exclusion list of activities.

(ii) Safe access to Roof for installation and operation through life cycle of facility

Level of Concern: Moderate

Mitigation Measures: None required, if access to roof is provided through existing staircase on 24 X 365 days (including holidays and Sundays).

In case, external access dedicated to GRPV must be installed, it shall then have adequate safe access to roof for installers and O&M personnel e.g. staircase slope, riser height, tread width, landings, hand railing etc.

(iii) Roof Condition and Drainage

Level of Concern: Moderate

1. _____

[16] PM Surya Ghar: Muft Bijli Yojna – Redefining Solar Power and Energy Access: Press Information Bureau, Government of India.

Mitigation Measures: None required, in case roof is in good condition with no leak/cracks and satisfactory drainage. Permissions for making good the roof and carry out rectifications to ensure roof has a proper and adequate drainage shall be obtained from the building owner, if required.

(iv) Roof Material

Level of Concern: High

Mitigation Measures: Potential investments will not be done on roofs with potential carcinogenic material (such as asbestos). Such sites will not be implemented through the Program and will be included in exclusion list of activities.

(v) Lopping and Pruning of Trees for shadow free areas on roof and ground mounted panels

Level of Concern: Moderate

Mitigation Measures: None required, if there are no trees adjacent to building. In case of trees, which cast shadow on the roof or proposed site for elevate ground mounted PV installation, permissions from competent authorities are to be obtained for cutting or periodic lopping and/or pruning of trees through life cycle of facility, as maybe the case.

(vi) Availability of Water to clean panels off accumulated dues, dirt and pollution to maintain their operational efficiency.

Level of Concern: Moderate

Accumulated dust, dirt, and pollution on panels can reduce energy output by up to 20–30%. The frequency of cleaning depends on environmental factors such as dust levels, rainfall, and local air pollution, which vary significantly across different regions in India. On an average, a residential solar PV system of 3–5 kW capacity might consume approximately 50–100 liters of water per cleaning session. In arid and semi-arid regions, such as Rajasthan and parts of Gujarat, where dust accumulation is high, cleaning may be required monthly or even more frequently during dry seasons. In contrast, regions with higher rainfall, such as Kerala or coastal Maharashtra, may require cleaning only every 2–3 months. With cleaning conducted 6–12 times a year, annual water consumption per household for panel cleaning could range from 300 to 1,200 liters.

Mitigation Measures: None required, if assured dedicated extension from existing water supply system to the building. If not, services in terms of a new municipal water supply connection or commercial water tankers are to be availed. If commercial water tankers are not viable or not dependable then a new tube well must be installed to serve the washing/cleaning needs of the panels. Required permissions for a new tube well from competent authorities and consents from building owner are to be obtained, as may be required. Water-efficient and sustainable cleaning practices, such as using low-pressure water jets, waterless cleaning systems, or automated robotic cleaning technologies that require minimal or no water (in case of larger footprint installations) can be promoted through the Program. Additionally, community awareness campaigns should be undertaken to educate residents on effective cleaning methods and the importance of water conservation in the context of solar PV maintenance.

(vii) Disposal of damaged and/or discarded panels (Solar Waste)

Level of Concern: Moderate

Solar waste refers to both discarded modules as well as scrap generated during the cell and module manufacturing processes. Modules can be discarded either when solar PV modules reach the end of their

functional life or upon sustaining damages from activities such as transportation, handling and installation.[17] Solar waste in India should be treated appropriately according to the Electronic Waste Management Rules 2022. If not managed appropriately, opportunity to reclaim valuable minerals is lost as well as these panels may contribute to electronic waste, posing potential environmental and health risks due to the presence of hazardous materials such as lead and cadmium. In the absence of strict enforcement, some panels end up in landfills or informal recycling units, leading to environmental hazards. As per the E-Waste (Management) Rules, 2022, under the Extended Producer Responsibility (EPR) framework, manufacturers and producers pf PV cells and panels are required to establish collection and recycling mechanisms for solar photovoltaic (PV) waste. Chapter V of the rules mandates that every manufacturer and producer of solar PV modules, panels, or cells must store solar PV waste generated until 2034–2035 as per guidelines issued by the Central Pollution Control Board (CPCB). Additionally, manufacturers must maintain an inventory of solar PV waste on a designated portal and comply with storage and record-keeping requirements. However, storing and managing such large quantities of solar waste for extended periods presents a significant challenge for manufacturers.

As per prevalent practices and experience, it is calculated that PMSG is expected to contribute approximately 2 million MT of waste over the next 20–25 years. This is based on certain assumptions, including that average installation over 10 million households will be 4 kW per household. As per CEEW's study, India's installed 66.7 gigawatt (GW) capacity, as of FY23, has generated about 0.1 MT of cumulative waste, which will increase to 0.34 MT by 2030. Around 67 per cent of this waste is expected to be generated in five states: Rajasthan, Gujarat, Karnataka, Tamil Nadu and Andhra Pradesh. This solar waste in India is expected to reach 600,000 metric tons (MT) by 2030 from existing and new capacities. By 2050, cumulative solar waste is projected to rise to 19 million MT. Further details are included in Annexure 1.

Mitigation Measures: It is essential to provide enabling environment to stakeholders, including manufacturers, Ministry of Environment Forest and Climate Change (MoEFCC) and MNRE, to adhere to E-Waste (Management) Rules 2022.

(viii) Safety of installers and O&M personnel

Level of Concern: Moderate

Mitigation Measures: Installers and O&M personnel should be provided with a personal fall protection system, ensuring proper lifting and ladder usage policies and procedures at sites, and installation of the guard rail system on a rooftop during the construction phase.

(ix) Safety and Fire Hazards

Level of Concern: Moderate

Mitigation Measures: Checks and balances for risks related to electrical hazards to be adhered during implementation of Program. Along with a fire protection and extinguishing system, basic measures shall be provided such as protection of equipment with ground fault circuit interrupters. A short circuit protection system shall be installed. Additionally, community awareness campaigns should be undertaken to educate residents to manage associated risks.

1.

[17] Council with Energy, Environment and Water's report on Enabling a Circular Economy in India's Solar Industry - https://www.ceew.in/publications/how-can-india-enable-circular-economy-with-solar-energy-waste-management-disposal

Social Effects

The project will bring in overall positive impacts to beneficiary households. The specific positive impacts are:

- lower-cost electricity for participating households
- employment opportunities as part of the installations under the program
- education and thereby employment opportunities to children due to assured power supply

The risks are listed below:

- social exclusion of low-electricity consumption households as they are not be able to access due to a) Low awareness about the RTS PV, b) Difficulties in registering on the website, c) Absence of suitable space for installation/ technical issues with available space, d) Delays in feasibility studies, e) Delays in subsidy payments, f) Lack of technical skills to operate and maintain, g) No/inadequate training on RTS PV.
- Non-compliance with state and national labor welfare measures, such as labour license, labour registration, labour insurance, equal wages, grievance redressal, etc.
- Labour health and safety during solar panel installation and operation
- Community health and safety during solar panel installation and operation
- As it is mostly men working on the installation, women safety during installation of the solar panels;
- gaps in the system to track grievances and feedback from the customers/ beneficiaries.
- For ground-mounted solar, if government/ community land is used, the risk of loss of land for any particular use, livelihood loss or loss of access/ access restrictions due to solar installations on the government/ community land.

CHAPTER 4: REGULATORY FRAMEWORK REVIEW

Introduction

1. The PM Surya Ghar: Muft Bijli Yojana will operate under a comprehensive governance framework involving national and state-level policies that are specifically designed for rooftop solar energy systems. These policies will provide a structured approach to ensuring the environmental, social, and technical integrity of the program. A detailed list of the relevant policies, regulations, and laws is provided, which outlines the key legislative measures applicable at both the national and state levels. These include policies on solar energy development, environmental protection, which will guide the implementation of residential rooftop solar projects.

2. The program will adhere to the provisions of national frameworks such as the National Electricity Policy (NEP), the National Action Plan on Climate Change (NAPCC), and the National Solar Mission (NSM), which aim to enhance the adoption of renewable energy while reducing the environmental impact. The overarching goal is to integrate grid connected rooftop solar systems into India's energy mix while addressing the country's climate goals.

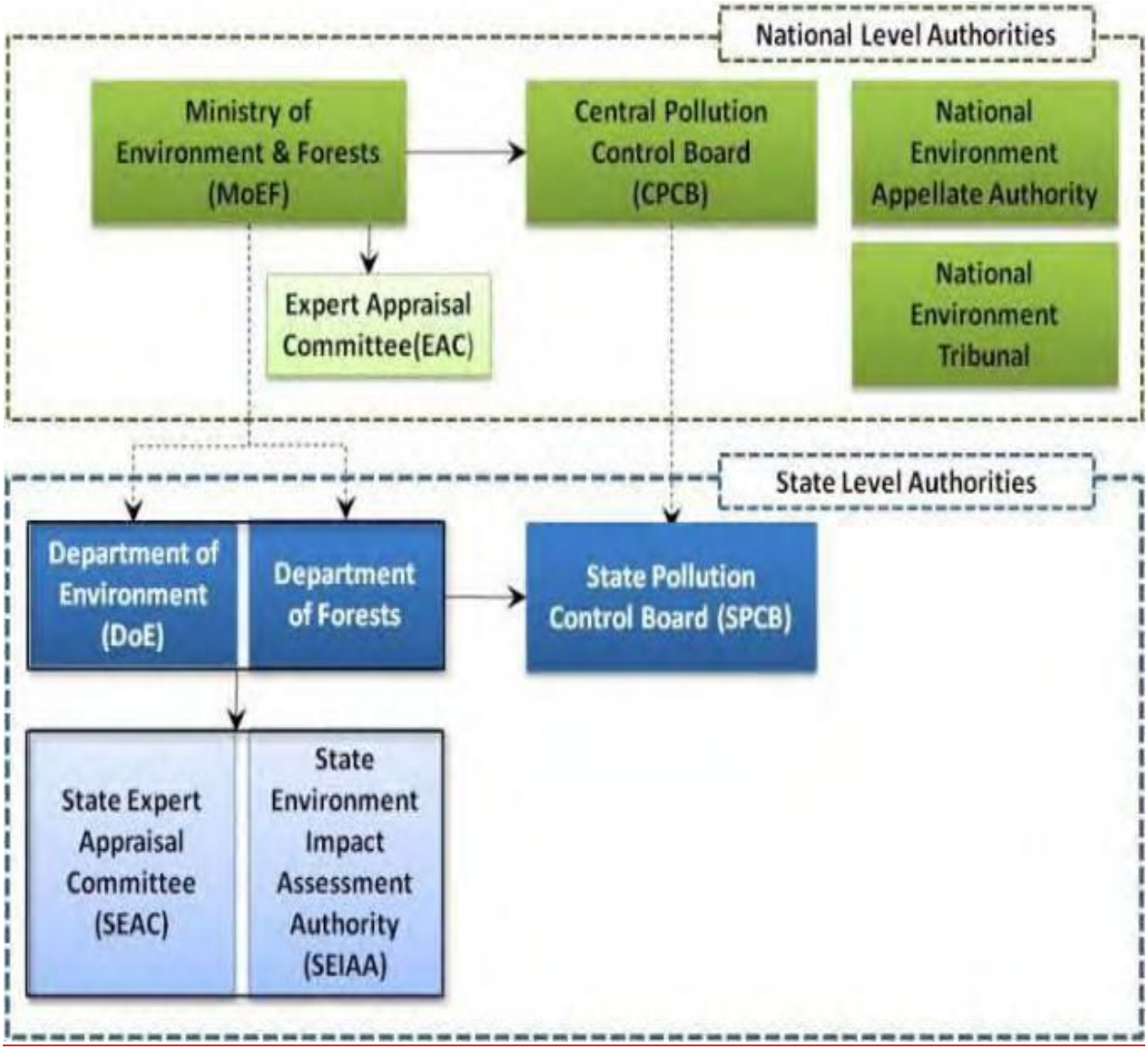
3. The PM Surya Ghar scheme will also comply with the Environment Protection Act, 1986, which provide the necessary legal framework for managing environmental risks, such as waste disposal, water use, and biodiversity concerns.

Applicable National Laws and Regulations

4. In the context of Environment regulations, the Government of India through the Ministry of Environment Forests and Climate Change (MoEFCC), is responsible for environmental policy and regulatory formulation, as well as overseeing implementation of these regulations. The State Governments are responsible for ensuring implementation and enforcement of national and state environmental legislations. The Central Pollution Control Board at the central level, along with its counterparts, the State Pollution Control Boards/Committees, are jointly responsible for implementation of legislations related to prevention and control of environmental pollution.

5. The institutional structure for environmental policy and regulatory framework in India is given in **Figure 3**.

Figure 3: Institutional Set-up for Environmental Policy and Regulatory Framework



6. The key environmental regulations at the National, State and local levels, which are relevant and applicable for GRPV, including elevated ground mounted PV installations, and compliance requirements, are summarized in Table 6.

Table 6: Environmental Regulations and their applicability to PMSG Program

S. No.	Environmental Acts and Regulations	Requirement of the Regulations	Applicability to the GRPV Facility
National Level			
(i)	Environmental Impact Assessment Notification, 2006 and latest amendments	Obtaining prior Environmental Clearance from Ministry of Environment and Forests (MoEF) or State Environmental Impact Assessment Authority (SEIAA) for new/expansion of project or activities	Not applicable to the Project. Solar Power Projects including Rooftop projects are exempted from purview of EIA notification, 2006 as

S. No.	Environmental Acts and Regulations	Requirement of the Regulations	Applicability to the GRPV Facility
		listed in the Schedule of EIA Notification 2006.	per circular issued by the MoEF, GoI.
State Level			
(ii)	Water (Prevention and Control of Pollution) Act, 1974 and amendments thereof	Obtaining Consent for Establishing (CFE) before commencement of construction and Consent for Operation (CFO) are to be obtained prior to start of construction and before commissioning of Project from Pollution Control Board (PCB).	Applicable as required by the respective jurisdiction within whose jurisdiction the respective sub-project lies. Consent for Establish (CFE) at the time of construction and Consent for Operation (CFO) at the time of commission of the plant is to be obtained from the respective Pollution Control Boards.
(iii)	Air (Prevention and Control of Pollution) Act, 1981 and amendments	Obtaining Consent for Establishing (CFE) before commencement of construction and Consent for Operation (CFO) before commissioning from PCB.	As above.
(iv)	Hazardous Waste (Management and Handling) Rules, 2008 and amendments thereafter	Obtaining authorization from PCB for management and handling of Hazardous Waste like discarded/damaged PV panels, lube and/or transformer oils etc. during installation and operation phases.	Authorization from the State Pollution Control Board shall be obtained for lube and/or transformer oils etc. during installation and operation phases, as may be applicable.
(v)	E-Waste (Management) Rules, 2022	Obtaining authorization from PCB for management and handling of E-Waste like discarded/damaged PV panels, lube and/or transformer oils etc. during installation and operation phases.	Authorization from the State Pollution Control Board shall be obtained for solar waste during installation and operation phases, as may be applicable.
Local Level			
(vi)	Tree cutting, lopping and pruning	Permission has to be sought from the designated local tree authority designated by the Department of Environment and Forests.	Authorization from Department of Environment and Forests shall be obtained as necessary.
(vii)	Disposal of construction spoil and debris	Permission of local self-government bodies (ULBs) will be required for disposal of construction spoil and debris.	To be taken, if required.

S. No.	Environmental Acts and Regulations	Requirement of the Regulations	Applicability to the GRPV Facility
Health and Safety Acts and Rules			
(viii)	Building and Other construction Workers (Regulation of Employment and Conditions of Service) Act, 1996	Regulates the employment and conditions of service of building and other construction workers and to provide for their safety, health and welfare.	If required, project proponent shall ensure compliance of the requirements under the Act during construction.
(ix)	Petroleum Act, 1934 and Rules 2002	Regulates import, transport, storage, production, refining and blending of petroleum as approval is to be sought from the Chief Inspectorate of Explosives for storage of diesel.	Not applicable.
(x)	Indian Electricity Act 2003 and Rules 1956 amended to 2000	The safety requirements specified in Chapter IV of the Act are to be observed.	Project proponent shall ensure compliance of the requirements under the Act during construction, installation and Operation phases.
(xi)	World Bank Group Environmental, Health and Safety Guidelines (EHSG)	Provides the good practice that should be reviewed and followed for Bank supported projects. Relevant guidelines are those covering General Aspects, and Transmission and Distribution Sectoral guidelines.	Project proponent shall confirm that WB group's EHSG guidelines applicable to rooftop solar will be followed during installation and operation of plants as reflected in the Annexure 5.

Social Acts and Regulations:

7. The following are the social related regulation relevant to the program:

a) Right to Information Act, 2005: The Act provides for setting out the practical regime of right to information for citizens to secure access to information under the control of public authorities, in order to promote transparency and accountability in the working of every public authority, the constitution of a Central Information Commission and State Information Commissions and for matters connected therewith or incidental thereto. Under this act, the information seeker need to pay for the charges, towards cost of paper, printing and preparation. All documents pertaining to the P4R would need to be disclosed to public, to avoid costs to the information seekers.

b) Article 366 (25) of the Constitution of India Article 244(1) of Constitution of India - The Fifth Schedule under Article 244(1) of a subsequent Act of Constitution "Scheduled Areas" as such areas as the President may by order declare to be Scheduled Areas after consultation with Governor of that State: The Fifth Schedule of the Constitution of India deals with administration and control of

scheduled areas and scheduled tribes in these areas. Defines following essential characteristics, for a community to be identified as Scheduled Tribes are;

a. Indications of primitive traits;
b. Distinctive culture;
c. Shyness of contact with the community at large;
d. Geographical isolation; and
e. Backwardness.
f. The criteria for declaring any area as a "Scheduled Area" under the Fifth Schedule are; (a) preponderance of tribal population, (b) compactness and reasonable size of the area, (c) a viable administrative entity such as a district, block or Taluka, and (d) economic backwardness of the area as compared to the neighboring areas. Fully applicable to the P4R Program. Applicable when tribals are present.

c) Scheduled Tribes and Other Traditional Forest Dwellers (Recognition of Forest Rights) Act, 2006: To recognize and vest the forest rights and occupation in forest land in forest dwelling STs and other traditional forest dwellers who are residing in such forests for generations but whose rights could not be recorded. Its objective is to facilitate the overall development and welfare of the tribal people by empowering them socially, economically, politically without any impact on their culture, habitation and tradition and in terms of their age-old rights and privileges. Fully applicable to the P4R Program. Applicable when tribals are present

d) Panchayats (Extension to the Scheduled Areas) Act, 1996: The Gram Sabha or the Panchayats at the appropriate level shall be consulted before making the acquisition of land in the Scheduled Areas for development projects and before re-settling or rehabilitating persons affected by such projects in the Scheduled Areas. Fully applicable to the P4R Program activities in Schedule V and VI areas.

e) Building and Other construction Workers' (Regulation of employment and conditions of service) Act, 1996 and corresponding States' Rules: All the establishments, which carry on building or other construction work and employ 10 or more workers are covered under this Act. Employer of the establishment is required to provide health and safety measures at the building or construction work and other welfare measures, such as canteens, first-aid facilities, ambulance, housing accommodation for workers near the workplace, among other benefits under the Rules. Every building worker within age limit of 18 to 60 years, who has worked for not less than 90 days in the preceding year is shall eligible for registration as a beneficiary under this act. Fully applicable to the PforR Program.

f) Building and other construction workers Welfare Cess Act, 1996 and corresponding States' Rules: Provides for levy and collection of a cess on the cost of construction incurred by employers to augmenting the resources of the Building and Other construction Workers' welfare Board constituted under this Act. Fully applicable to the P4R Program.

g) The Child Labour (Prohibition & Regulation) Amendment Act, 2016 and corresponding States' Rules: This Act prohibits employment of children below 14 years of age in certain occupations and provides for regulation of employment of children in all other occupations and processes. Employment of child labour is prohibited in Building and construction industry. There should not be any child labour (less than 14 years) in any project activity and adolescents (above 14 and less than 18 years) in any hazardous activity. Fully applicable to the P4R Program.

h) Payment of Wages Act, 1936, The Minimum Wages Act 1948 and corresponding States' Rules:

Lays down as to by what date, wages are to be paid, when it will be paid and what deductions be made from the wages of the workers, if any. Fully applicable to the P4R Program.

i) Payment of Gratuity Act, 1972 and corresponding States' Rules: Gratuity is payable to an employee under the Act on satisfaction of certain conditions on separation, if an employee has completed 5 years of service with employer. Fully applicable to the P4R Program.

j) Employees Provident Fund and Miscellaneous Provision Act, 1952 and corresponding States' Rules: Every establishment, which engages in any industry specified under schedule 1 and in which 20 or more persons are employed are under the purview of this Act. Provides for monthly contributions by the employer and as well as by workers with a provision as return of pension of a lump sum (principal and interest accrued) at the end of his/her service term). Fully applicable to the P4R Program.

k) Maternity Benefit Act, 1961 and corresponding States' Rules: Provides for maternity leave for women, during pregnancy and after giving birth and some other benefits to women employees, in case of medical recommendation of bed rest or miscarriage etc. Fully applicable to the P4R Program.

l) Contract Labour (Regulation and Abolition) Act, 1970, The Contract Labour (Regulation & Abolition) Act, 1970 and Rules and corresponding States' Rules: Provides for certain welfare measures to be mandatorily provided by the contractor to the contract labour. Fully applicable to the P4R Program.

m) Equal Remuneration Act, 1979 and corresponding States' Rules: Provides for payment of equal wages for equivalent work to male and female workers without any discrimination against women. Fully applicable to the P4R Program

n) The Bonded Labour (Abolition) Act 1976, Bonded Labour System (Abolition) Rules 1976 and corresponding States' Rules: An Act to provide for the abolition of bonded labour system, with a view to prevent economic and physical exploitation of the weaker sections of the people and for all matters connected there with or incidental thereto. Fully applicable to the P4R Program.

o) The Trade Union Act, 1926 and corresponding States' Rules: Lays down the procedure for registration of trade union of workers and employers. The trade unions registered under the Act have been given certain immunities for civil and criminal liabilities. Fully applicable to the P4R Program.

p) Workmen's Compensation Act, 1923 and corresponding States' Rules: Provides for compensation in case of injury by accident arising out of and during the course of employment. Fully applicable to the P4R Program.

q) ESI Act, 1948 (Employees State Insurance Act, 1948) and corresponding States' Rules: Employees State Insurance Act provides for health care and hospitalization benefits for construction work force. Fully applicable to the P4R Program.

r) Inter-state Migrant Workmen's (Regulation of Employment and Conditions of Service) Act, 1979 and corresponding States' Rules: The inter-state migrant workers, in an establishment to which the Act becomes applicable, are required to be provided with certain facilities such as housing, medical aid, traveling expenses from home to the establishment and back etc. Fully applicable to the P4R

Program.

s) National Policy on Safety, Health and Environment at Workplace (NPSHEW), 2009: The GoI's National Policy seeks continuous improvement in Occupational Safety, Health and Working Conditions at workplaces. The policy objectives are to achieve: -(a) Reduction in incidence of work-related injuries, fatalities, diseases, disasters, and loss of national assets; (b) Comprehensive data base for facilitating better performance and monitoring; (c) Enhancement of community awareness regarding safety, health and environment at workplace related areas; (d) Continually increasing community expectation of workplace health and safety standards and (e) Improving safety, health and environment at workplace by creation of "green jobs" contributing to sustainable enterprise development. India has currently consolidated all labour laws into four comprehensive Labour Codes which regulate: (i) Wages, (ii) Industrial Relations, (iii) Social Security, and (iv) Occupational Safety, Health and Working Conditions. These Codes have been passed by Parliament and have received Presidential Assent. Draft National Rules for operationalising the Codes have been circulated for comments but have not been passed yet. Since labour falls under the Concurrent list, both, the central and state governments are empowered to legislate on the subject. Thus, for implementation of these Codes, Rules under these must be notified by central as well as the states for enforcing these in the state. The last date for finalizing the rules is March 2025. For now, the following national and state labour legislation, with reference to terms and conditions of work, compensation and benefits are summarised in this table, will still be applicable for this P4R. Fully applicable to the P4R Program.

t) The National Policy for the Empowerment of Women, 2001: The National Policy for the Empowerment of Women adopted in 2001 states that all forms of violence against women, physical and mental, whether at domestic or societal levels, including those arising from customs, traditions or accepted practices shall be dealt with effectively with a view to eliminate its incidence. Institutions and mechanisms/schemes for assistance will be created and strengthened for prevention of such violence, including sexual harassment at work-place and customs like dowry; for the rehabilitation of the victims of violence and for taking effective action against the perpetrators of such violence. The Draft National Policy for Women in 2016 was an attempt to further the mission of equal rights and opportunities for women in family, community, workplace and governance. India has signed and ratified Convention on Elimination of Discrimination against Women (CEDAW). Since then, the National policy for Women and other policies and amendments on acts has been reflecting the principles highlighted in the related international conventions. The Convention on the Rights of the Children lays out a universal definition of 'child'. It articulates how children should be treated in a non-discriminatory manner and calls for action that considers the best interests of the child. Fully applicable to the P4R Program.

u) Sexual Harassment at the Workplace (Prevention, Prohibition and Redressal) Act, 2013 and amendments: The act provides for protection against sexual harassment of women at workplace and for the prevention and redressal of complaints of sexual harassment and for matters connected therewith or incidental thereto. Fully applicable to the P4R Program.

v) The Protection of Women from Domestic Violence Act, 2005: This Act defines domestic violence, describes the powers and duties of protection officers, service providers and lists the procedures for obtain reliefs. Fully applicable to the P4R Program.

w) The Protection of Children from Sexual Offences Act, 2012: This act has been framed to strengthen the legal measures for protecting children from any kind of sexual offences. Despite having sexual offences covered under different sections of Indian Penal Code, those existing sections do not

entirely cover similar offences against children. Recognizing the children as a person of 18 years or below, the various offences have been clearly defined under this Act. The Act also defines the gravity of certain offences carried out against children and suggests stringent punishments accordingly. Fully applicable to the P4R Program.

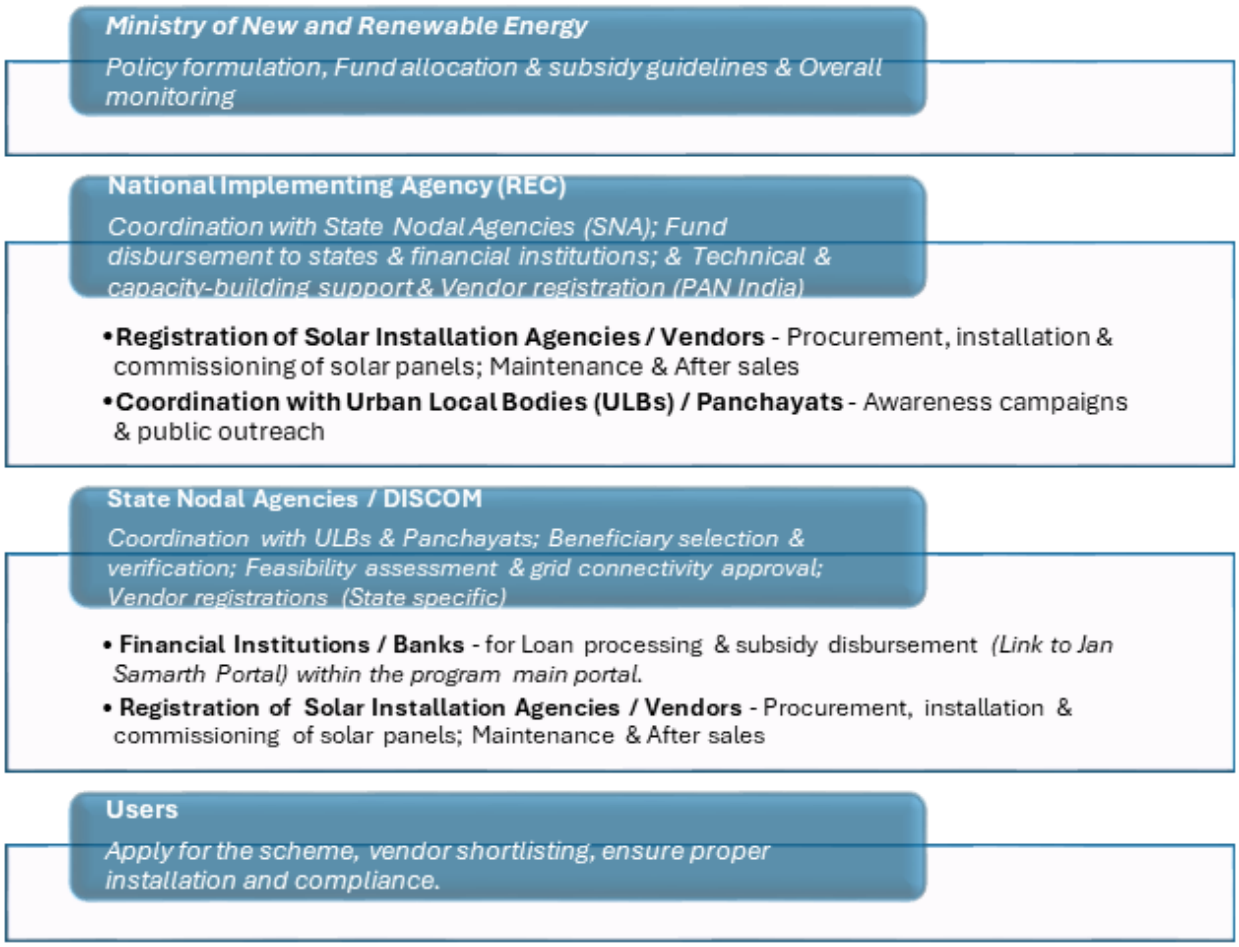
x) The Immoral Traffic (Prevention) Act in 1956: India ratified the Convention for the Suppression of the Traffic in Persons and of the Exploitation of the Prostitution of Others and following that, the country enacted The Immoral Traffic (Prevention) Act in 1956 and punishes trafficking in relation to prostitution. Fully applicable to the P4R Program.

y) Guidelines/SOPs related to Covid-19 like pandemics outbreaks, Standard Operating Procedures (SOPs) and Guidelines for Construction Sites for COVID-19 like outbreaks issued by Government of India/ States: These Guidelines cover aspects relating to safety measures for workers, machineries, tools, and Emergency protocol in case of detection of symptoms of COVID 19 like outbreaks to be observed by Project Manager of Contractor. The orders giving instructions relating to attendance, thermal scanning, no-deduction of wage; holding of meetings; frequent sanitation of workspaces, etc. need to be followed. Fully applicable to the P4R Program.

CHAPTER 5: PROGRAM CAPACITY ASSESSMENT

Institutional Arrangement

1. The government program aims to increase penetration of solar PV modules in the country through PMSG. Below flow chart provides a brief of the key agencies involved in the implementation of the government program.



2. The roles and responsibilities of the various agencies involved in the implementation are as follows:

(i) **Ministry of New and Renewable Energy (MNRE):** MNRE is the key agency for the implementation of the government program. The program is being implemented through a dedicated web portal and a mobile application. MNRE develops the implementation framework, setting guidelines, technical specifications, and quality standards to ensure uniformity and safety in installations across the country. All monitoring will be centralized at the Ministry level. The Ministry also plays a coordinating role, working closely with state agencies, distribution companies (DISCOMs), financial institutions, and other stakeholders to facilitate smooth execution and resolve operational challenges.

(ii) **REC Limited*:*** REC acts as the National Program Implementation Agency (NPIA) and supports MNRE in implementing and monitoring the program through management of funds for the program. REC Limited is responsible for disbursement of funds under different scheme heads and interact with different stakeholders on a regular basis. Every component will be directly or indirectly associated with REC since they will be interacting with each state and discoms. REC is also responsible for the vendor registration

particularly for the PAN India level wherein a vendor submits a declaration along with a bank guarantee of 25 lakhs for registration.

(iii) **State Level Agency (DISCOMs):** DISCOMs are responsible for consumer registration and application processing, assessing feasibility and granting grid connectivity approval, providing technical clearance for integration, reporting implementation progress to REC and MNRE, and ensuring compliance with quality and safety standards through two physical verifications during installation and commissioning. DISCOMs are also responsible for registering vendors/RESCOs/ULAs if they are operating in a single state or DISCOM jurisdiction.

(iv) **End User / Beneficiary:** The residential consumer may be an individual household or a group housing society or a residents welfare association. The consumer will be responsible for registering himself with the web portal to avail the program benefits. Consumers are responsible for taking necessary approvals/clearances from concerned authorities and ensuring compliance with technical standards and maintenance for at least five years.

(v) **Vendors/RESCOs/ULAs**:

- Vendors are independent firms specializing in installation of PV modules. They primarily work through channel partners, including those from outside the solar PV sector, who are interested in supporting the program despite lacking prior experience. Currently, there are about 12,000 solar channel partners; out of these 90% are new entrant with potential for becoming partners. The vendors are required to be registered either with MNRE or DISCOMs and are selected by the consumer to undertake installation in their premises.
- RESCO is a third-party to finance, install, own, and maintain rooftop solar systems for at least five years, with consumers paying only for the electricity consumed. Ownership may transfer to the consumer after the contract period, and surplus power can be sold to DISCOMs through power purchase agreements. DISCOMs facilitate tariff collection and subsidy disbursal, while a Payment Security Mechanism (PSM) ensures timely payments to RESCOs. In contrast, the Utility Led Aggregation (ULA) Model enables DISCOMs or state entities to install and own solar systems on behalf of consumers, with two variations - Utility-Owned Assets, where ownership remains with the utility for at least five years before potential transfer, and Consumer-Owned Assets, where consumers own the system from commissioning, contributing up to 10% of costs.
- The vendors/RESCOs/ULAs are responsible for the procurement & supply of solar equipment; installation & commissioning (conducting site assessments and designing solar system layouts); coordinating with DISCOMs for grid connection and net metering activation and ensuring installations meet government regulations, safety codes, and performance benchmarks. These entities are also responsible for providing post-installation support, including troubleshooting and system maintenance during the mandatory 5-year O&M stage.

3. **Monitoring:** MNRE is setting up a multi-tier mechanism for monitoring the implementation of the government program. A Group of Ministers will be constituted for guidance and coordination with states. A Steering Committee under the chairmanship of the Cabinet Secretary and including Secretaries of the Ministry of New and Renewable Energy, the Department of Financial Services, the Ministry of Power, the Ministry of Housing and Urban Affairs, and the Ministry of Rural Development will monitor the implementation of the scheme. There will be a State Level Coordination Committee headed by the Chief Secretary, Advisor to the Lieutenant Governor or Administrator, and comprising Secretaries of key departments to coordinate and monitor the PMSG program implementation in the State/Union Territory.

There will be a District Level Committee headed by the District Magistrate or District Collector and comprising other members to monitor the implementation of the PMSG program at the district level.[18]

4. **Capacity Building:** The Program will support the government program's capacity building efforts to create a skilled workforce for efficient rooftop solar installations. PMSG aims to impart training on installation, operation, maintenance, and repair of solar rooftop systems, targeting electricians, engineers, vendors, and DISCOM officials. Additionally, stakeholders such as financial institutions, state renewable agencies, and government departments will receive capacity-building support to ensure effective scheme implementation.

5. Implementation will be structured at national and state levels, with the NPIA formulating the National Skilling and Capacity Building Action Plan (NSCBAP) and SIAs developing State Skilling and Capacity Building Action Plans (SSCBAPs). Partner Ministries will also contribute through Departmental Skilling and Capacity Building Action Plans (DSCBAPs). The program will be integrated with the Skill India Digital Hub (SIDH) platform for monitoring and evaluation.

6. **Awareness and Outreach**: The Program will support PMSG to generate awareness regarding widespread adoption of rooftop solar installations. PMSG aims to promote informed decision making by consumers regarding installation processes, technical aspects, and best practices

7. Implementation is structured at national, state, and departmental levels, with the NPIA overseeing the National Awareness & Outreach Plan (NAOP), SIAs managing State Awareness & Outreach Plans (SAOPs), and partner ministries formulating Departmental Awareness & Outreach Plans (DAOPs). Collaborative efforts with Self-Help Groups, Urban Local Bodies, Panchayati Raj Institutions, and Common Service Centers (CSCs) will further amplify outreach efforts.

Sub-Project Appraisal/Due Diligence

It is yet to understood from MNRE how the proposals are appraised to ensure environmental and social aspects are accounted. Following is an example from ongoing Program on Grid Connected Rooftop Solar with SBI taken from the Operations Manual of SBI:

Table 7: Stage wise Due-diligence process

Stage	Environment and Social aspects considered
At the initial stage (for project under Project mode or assessment of individual projects in Program mode)	• Statutory Clearances - Pollution Clearance, Project Clearance and other Approvals / Clearances / Permissions wherever applicable to be obtained as per the implementation stage of the project. • Availability of critical infrastructure- position of power evacuation infrastructure, availability of water etc. • Rooftop Solar Power Project - (i) CAPEX Model - building should be owned by developer. (ii) Third Party Model (RESCO Model/Aggregator Model) – Draft of the lease agreement for rooftop with access rights, with owner of building should be examined. • Site visit to be conducted and discreet enquiries to be carried out/landmark to be noted. • Social o Training and Skill Development plans implementation o AOP implementation o Roof rights of the consumer o Proper Contract with Empaneled Agencies o Land availability for ground mounted installations

1.

[18] Operational guidelines for implementation of PM-Surya Ghar: Muft Bijli Yojana for the component "Central Financial Assistance to Residential Consumers," MNRE, GoI, June 7, 2024.

Stage	Environment and Social aspects considered
	○ Mandatory consultations in tribal areas ○ Community health and safety ○ Ease in operating and navigating the portal
Pre-approval credit process (for projects under the Project mode (i.e. single projects) or assessment of individual projects in the Program mode, i.e. aggregated projects)	Technical and Contractual Feasibility • Rooftop ○ Ownership – the ownership details (with a copy of ownership documents) of the rooftop where project is proposed to be installed. In case the roof is being leased / rented or to be leased/rented, the lease contract/draft or the rental agreement should be obtained and is to be analyzed for the value of the lease/ rental, the tenor of lease, the risk of early termination etc. ○ Ease of accessing the rooftop during construction and also during the operation & maintenance period. • Rooftop profiling and site assessment to analyze aggregate rooftop area, rooftop area suitable for PV project development, potential for PV capacity etc. • Other Infrastructure ○ Water requirement and its source of supply, for cleaning of panels. • Approvals and clearances ○ Copies of all approvals and clearances are required to be obtained before implementation and analyzed for their validity. A plan for obtaining other approvals and clearances (that are required at later stages) is also to be obtained. • Environment, Health and Safety norms Assessment ○ A copy of the plan / manual for compliance with the agreed Environment Health Social and Safety (EHSS) norms, including fire safety measures adopted during construction and post-commissioning, is to be obtained and analyzed. ○ Assessing project compliance with the SBI's environmental and social requirements. ○ Site audits are to be conducted during the implementation and operations period by a lender's independent engineer (LIE) or by SBI's staff (depending on the size of the investment as specified in the guidelines) in order to assess on-site practices and to ensure compliance with documented and agreed procedures. • Social Assessment ○ Labour compliances by Empanelled Agencies ○ Ease in reimbursing the CFA subsidy amount ○ Credit Worthiness of the Consumer/ CIBIL Score ○ Overall Interest rate including hidden charges
Assessment	EHSS requirements shall be discussed in the loan proposal. • Social: Benefit Cost Ratio for the consumer (considering capital and operational costs)
Approval	• A draft or a copy of the tripartite agreement/lease agreement among rooftop owner, aggregator and SBI should be provided, wherever applicable. • Licenses/approval from regulatory authorities - Environmental Clearances, Project approval from State Agency for setting up the plant, No Objection Certificate (NoC) from the Pollution Control Board, and/or a Grid Connectivity Agreement with the authority concerned etc.
Disbursement	• Ensure that all the licenses, approvals, agreements with third parties, as required during implementation, have been put in place and all the pre-disbursement conditions of loan approval have been fulfilled or that the sub-loan applicant has obtained written approval from a qualified authority for any deviation. • Social: Reimbursement of Subsidy without delays

Stage	Environment and Social aspects considered
Post approval credit process (*Follow up, Supervision and Monitoring)*	• Ensure that the end-use of funds is consistent with intended purposes. • Detect any deviation from terms of the loan approval. • Make periodic assessments of the health of the loan advances by calculating and documenting some of the key performance indicators like profitability, activity level, and management of the unit. Follow up will also be undertaken to ensure that the assets created are effectively utilized for productive purposes and are well maintained. • Identify early warning signals, and undertake review and reporting and initiate remedial measures proactively, thereby averting the incidence of incipient deterioration of financial health of the sub-borrower. • Ensure compliance with all internal and external reporting requirements covering the credit area. • Monitor, that the all the required approvals and clearances (as per the stage of implementation / operation) are obtained for the project (this will be done by SBI's staff or a lender's independent engineer, appointed by SBI). • Social o Labour compliances by Empaneled Agencies o GBV related SEA-SH o Proper maintenance by Empaneled Agency o Reimbursement of Subsidy o Grievance Redress by MNRE
Source: Operations Manual Grid-Connected Rooftop Solar Program, State Bank of India, 2015	

Assessment of Program Capacity

13. As per Electricity (Rights of Consumers) Rules 2020, required of technical feasibility report for GRPV upto 10 KW has been waived. MNRE has developed guidelines for installation of PV panels including design specifications for individual components. This also includes specifications for installations to be done in geographic areas impacted by high velocity winds. Electrical safety is ensured through certified components, DISCOM inspections, and vendor responsibilities, including a minimum 5-year maintenance agreement.

14. Currently, there is no provision of screening the proposal for sites located in sensitive areas including cultural heritage sites, protected areas, sensitive biodiversity areas, as well as for potentially carcinogenic roofing material.

15. Vendors/RESCOs/ULAs shall assess site feasibility and access issues. Any regulatory approvals required are responsibility of consumer. It is not clear how the system assesses whether these requirements have been met.

16. It is not clear whether the registration of vendors/RESCOs/ULAs factors their ISO certifications pertaining to ESHS.

17. Monitoring system of PMSG does not factor the environmental management aspects related to GRPV in its current form. The system may be enhanced with use of checklists and spatial mapping tools.

18. The Ministry of Skill Development and Entrepreneurship, under the Skilling India Program, has prepared aa training manual for "Rooftop Solar PV (installation and Maintenance)" under the NSQF Level 3 framework. The manual covers both theoretical and practical aspects regarding solar energy fundamentals, photovoltaic systems, installation procedures, maintenance, safety, and testing standards, aiming to skill technicians for the PMSG initiative. However, the manual does not cover emergency preparedness and response planning, incident reporting and protocols, stakeholder engagement, approvals

and necessary clearances, management of hazardous materials and solar waste, first aid provisions, availability of water for panel cleaning.

19. Currently, there is limited information available on the National Awareness & Outreach Plan (NAOP) and its implementation mechanisms. Given that the identified environmental risks are not accounted in the implementation and capacity building, it is assumed that these are not part of the outreach program as well.

20. The MNRE has various guidelines for the program, which are available on the website. MNRE does not have an Environment and Social Unit or Division to coordinate environmental and social management activities at an institutional level. All activities pertaining to environmental and social management are to be carried out state teams which include DISCOMs, Empaneled Agencies, who are assigned with implementation, commissioning and operation of RTS PV systems. The state level teams (which are yet to be assessed) appears to not have any social capacity to address the issues of inclusion, engagement, labor welfare, grievance redress, gender, etc.

21. The program is being implemented through the Portal. Several updates/ modifications are being carried out to the portal. There are several complaints about the solar portal; some of the key complaints are a) the portal is regularly down, b) subsidy payments are over delayed, c) the menu to download customer data are disabled, d) there are no replies from MNRE for the complaints, e) no feedback is collected from customers, f) registrations take a lot of time and consumers are unable to register, g) for filing any grivances the vendors and customers are unable to reach MNRE, etc.

22. As per the discussions at REC, the GRM Portal has received about 300,000 grievances out of which 274,000 are resolved. The GRM Portal has capacity to handle about 150 tickets a day. There are no provisions for escalation of grievances, resolution durations are not provided, the provision for feedback is not available. It is informed that the new GRM portal will have all these provisions.

23. As per the guidelines, CFA should have been processed within 15 days of approval by concerned DISOCM. The CAG Audit note mentions that there were 3314 beneficiaries waiting for 99 days and 3024 beneficiaries waiting for 100 or more days; the highest pendency for CFA is 390 days. The note says that no action is taken by the MNRE to resolve these issues. The note mentions that major chunk of project is implemented in four states and does not have pan India impact. The note further adds that there are no tracking and verification systems in the portal.

24. With regard to the IEC activities taken up by the program, the National Awareness and Outreach Plan (NAOP) and National Skilling and Capacity Building Plan (NSCBAP) for the Prime Minister Surya Ghar (PMSG), evaluation of the implementation of NAOP, etc. are yet to be received for assessment of awareness and outreach activities.

25. With regard to Capacity Building activities, the program aims to create about 300,000 skilled manpower for the project, out of which 100,000 will be Solar PV Technicians. The program aims to create awareness and motivate consumers to install Solar PV systems. The details of the trainings, trainees, outcome of the trainings, evaluation of the training programs, etc. are yet to be received for assessment.

26. The Key Implementing stakeholders are the Empaneled Agencies (about 10,000 Empaneled Agencies presently out of which about 20 were met during a virtual meeting of about an hour and a half) and DISCOMs at the state level. Discussions with the DISCOMs are yet to take place.

CHAPTER 6: ENVIRONMENTAL AND SOCIAL SYSTEMS ASSESSMENT

Introduction

1. According to the Bank Policy on Program-for-Results Financing, assessment and comparison of the core principles of Bank Policy against the country legal system for EHS and environmental and social management should be conducted. The purpose of core principles is to:

- promote environmental and social sustainability in the Program design; avoid, minimize, or mitigate adverse impacts, and promote informed decision-making relating to the Program's environmental and social impacts;
- avoid, minimize, or mitigate adverse impacts on natural habitats and physical cultural resources resulting from the Program;
- protect public and worker safety against the potential risks associated with: (i) construction and/or operations of facilities or other operational practices under the Program; (ii) exposure to toxic chemicals, hazardous wastes, and other dangerous materials under the Program; (iii) reconstruction or rehabilitation of infrastructure located in areas prone to natural hazards;
- manage land acquisition and loss of access to natural resources in a way that avoids or minimizes displacement, and assist the affected people in improving, or at the minimum restoring, their livelihoods and living standards;
- give due consideration to the cultural appropriateness of, and equitable access to, Program benefits, giving special attention to the rights and interests of the Indigenous Peoples and to the needs or concerns of vulnerable groups; and
- avoid exacerbating social conflict, especially in fragile states, post-conflict areas, or areas subject to territorial disputes.

E&S Systems Assessment

2. Analysis of the Program system against the core principles is as below.

Core Principle 1

Program E&S management systems are designed to (a) promote E&S sustainability in the Program design; (b) avoid, minimize, or mitigate adverse impacts; and (c) promote informed decision-making relating to a Program's E&S effects

Key Planning Elements:

3. Operate within an adequate legal and regulatory framework to guide Environmental and Social impact assessments at the Program level.

4. Incorporate recognized elements of environmental and social assessment good practice, including (a) early screening of potential effects; (b) consideration of strategic, technical, and site alternatives (including the "no action" alternative); (c) explicit assessment of potential induced, cumulative, and trans-boundary impacts; (d) identification of measures to mitigate adverse environmental or social impacts that cannot be otherwise avoided or minimized; (e) clear articulation of institutional responsibilities and resources to support implementation of plans; and (f) responsiveness and accountability through stakeholder consultation, timely dissemination of program information, and responsive grievance redress measures.

Existing System:

5. Under the existing system, the residential end user applies for the installation and subsidy for the solar rooftop system via a dedicated portal. Upon finalization of the vendor/RESCO/ULA by the end user,

the consumer provides rooftop access rights with the owner of the building. If any specific environmental clearances are required, the selected vendor/RESCO/ULA assists the end user in procurement of NOC / clearances if any.

Gaps:

6. The program focuses primarily on energy access, but early screening of environmental and social risks (e.g., siting of building in sensitive area, rooftop structural stability aspects, availability of water for cleaning, solar waste) are not comprehensively integrated at the planning stage.

7. Cumulative impacts of large-scale rooftop solar adoption (e.g., solar waste, impact on grid stability) and transboundary effects (e.g., solar waste and solar panel supply chain dependencies) are not explicitly addressed in the program design.

8. Implementation responsibility is distributed among state DISCOMs and central agencies, but there may be a lack of clarity on institutional roles in managing environmental and social safeguards, including capacity-building for local authorities, consumers, and vendors.

9. Monitoring system of PMSG does not factor the environmental management aspects related to GRPV in its current form.

Actions:

10. Adoption of WB group's EHSG guidelines applicable to GRPV (Annexure 5) as part of the system and ensuring that the relevant stakeholders are aware of and familiar with the requirements to ensure their implementation. MNRE shall make necessary arrangements to monitor and furnish reports with regards to these guidelines.

11. The implementation and monitoring system may be enhanced with use of checklists (Annexure 5) and spatial mapping tools.

12. Social: Currently MNRE manages Social Management issues, such as inclusion, engagement, labor welfare, grievance redress, gender aspects, etc., through the portal, as the link between the consumers and MNRE is only the portal, with Empaneled Agencies and DISCOMs physically present at the consumer's site during implementation. The present legal and regulatory frameworks at the national and state level, as detailed in the previous section, are fairly adequate and comparable with international standards. But, there are several gaps in compliance with the legal framework in terms of social management due to which quite a number of grievances are put up with MNRE. There are several complaints about the solar portal; some of the key complaints are a) the portal is regularly down, b) subsidy payments are over delayed, c) the menu to download customer data are disabled, d) there are no replies from MNRE for the complaints, e) no feedback is collected from customers, f) registrations take a lot of time and consumers are unable to register, g) for filing any grivances the vendors and customers are unable to reach MNRE, etc,.with There are no dedicated Social Management resources in terms of systems, plans, procedures and personnel. This requires a Social Management Systems and corresponding Plans detailing procedures and indicators for tracking and review.

Core Principle 2

Program E&S management systems are designed to avoid, minimize, or mitigate adverse impacts on natural habitats and physical cultural resources resulting from the Program. Program activities that involve the significant conversion or degradation of critical natural habitats or critical physical cultural heritage are not eligible for PforR financing

Key Planning Elements:

13. Includes appropriate measures for early identification and screening of potentially important biodiversity and cultural resource areas.

14. Supports and promotes the conservation, maintenance, and rehabilitation of natural habitats; avoids the significant conversion or degradation of critical natural habitats, and if avoiding the significant conversion of natural habitats is not technically feasible, includes measures to mitigate or offset impacts or program activities.

15. Takes into account potential adverse impacts on physical cultural property and as warranted, provides adequate measures to avoid, minimize, or mitigate such effects

Existing System:

16. The consumer, with support from vendor/RESCO/ULA submits the proposal with required documentation for clearances and permits.

Gaps:

17. The Guidelines does not provide guidance checklists or the parameters based on which proposals of consumers can be screened to be not impacting natural habitats or critical physical cultural heritage. Even in the National Portal there is no provision to screen the proposals submitted by the consumers to confirm if the installations encroach or degrade sensitive habitats, sensitive areas of biodiversity value, or are designated cultural heritage, are in areas protected as physical cultural resources/cultural heritage.

Actions:

18. PV installation on sites in protected areas are part of the Exclusion List of the Program. MNRE shall screen such sites and furnish periodic monitoring reports on the same.

Core Principle 3

Program E&S management systems are designed to protect public and worker safety against the potential risks associated with (a) the construction and/or operation of facilities or other operational practices under the Program; (b) exposure to toxic chemicals, hazardous wastes, and otherwise dangerous materials under the Program; and (c) reconstruction or rehabilitation of infrastructure located in areas prone to natural hazards

Key Planning Elements:

19. Promotes community, individual, and worker safety through the safe design, construction, operation, and maintenance of physical infrastructure, or in carrying out activities that may be dependent on such infrastructure with safety measures, inspections, or remedial works incorporated as needed.

20. Promotes the use of recognized good practice in the production, management, storage, transport, and disposal of hazardous materials generated through program construction or operations; and promotes the use of integrated pest management practices to manage or reduce pests or disease vectors; and provides training for workers involved in the production, procurement, storage, transport, use, and disposal of hazardous chemicals in accordance with international guidelines and conventions.

21. Includes measures to avoid, minimize, or mitigate community, individual, and worker risks when program activities are located within areas prone to natural hazards such as floods, hurricanes, earthquakes, or other severe weather or climate events.

Existing System:

22. MNRE requires E-Waste (Management) Rules, 2022 to be followed for management of solar waste whereby manufacturers and producers of PV panels and cells are required to establish collection and recycling mechanisms for solar photovoltaic (PV) waste. Chapter V of the Rules mandates that all such

manufacturers and producers must store solar waste until 2034–2035. Additionally, manufacturers must maintain an inventory of solar waste on a designated portal and comply with storage and record-keeping requirements.

23. The Guidelines includes only technical and design provisions for installation of PV panels in high wind velocity zones.

Gaps:

24. No known provisions exists in the Guidelines on environment, health and safety of communities and/or individuals for safe installation of infrastructure in normal conditions as well as in areas prone to extreme climate events.

25. Collection and storing large quantities of solar waste for extended period of time by manufacturers presents a significant challenge in terms of management since there are no guidelines on construction of these infrastructure and eventual management of solar waste.

Actions:

26. Vendor/RESCO/ULA registration process shall ensure that these firms, and their channel partners, are ISO certified for relevant and predefined EHS requirements.

27. MNRE, in association with Ministry of Environment Forests and Climate Change (MoEFCC), to develop enabling environment to manage solar waste effectively withing the span of the Program.

28. Social: The MNRE guidelines stipulates compliance with Central Electricy Authority's Notification of 08 June 2023 for electrical safety. The empaneled vendors response during the virtual meeting was mixed with regard to the safety practices; though they say they have PPE kits for the workers, with regard to safety practices some of them have their own manuals some do not. It is noted from their response that tests such vertigo test are not conducted on workers working at heights. There is a need for MNRE to develop a Safety (Public and Worker) Manual for the organization with regard to Solar Installations, particularly in view of solar installations on multistoried buildings.

Core Principle 4

Program E&S systems manage land acquisition and loss of access to natural resources in a way that avoids or minimizes displacement and assists affected people in improving, or at the minimum restoring, their livelihoods and living standards.

Social: MNRE rules out land acquisition for the program; the residential solar installations will be on consumers roof with roof rights. As the program includes ground mounted solar installation and community installations for those who do not have technically feasible roofs, government/ community land will be used for such installations, which poses a risk of loss of community land specified for various uses, livelihood loss and in some case risk of access restrictions. Such projects requiring government/ community land resulting in loss of community land for various uses, livelihood loss, access restrictions will be placed under exclusion list.

Core Principle 5

Program E&S systems give due consideration to the cultural appropriateness of, and equitable access to, Program benefits, giving special attention to the rights and interests of Indigenous Peoples/Sub-Saharan African Historically Underserved Traditional Local Communities, and to the needs or concerns of vulnerable groups

Social: The program has enhanced CFA for tribals. Presently, there are no recorded screening activities conducted by the implementing agencies identifying different property regimes, including common

property resources, customary or traditional rights to land or resource use, and the rights of tribals. The Implementing agencies do not have the required resources with necessary skills to undertake such assessments. The ESSA proposes that implementing agencies prepare, adopt and implement screening formats to identify different property regimes, including common property resources, customary or traditional rights to land or resource use, and the rights of tribals.

This aspect needs visits to Tribal areas where the program activities are taken up.

Core Principle 6

Program E&S systems avoid exacerbating social conflict, especially in fragile states, post-conflict areas, or areas subject to territorial disputes

Social: Any activities in fragile areas, and areas of conflicts/ disputes are placed under exclusion list.

CHAPTER 7: RECOMMENDATIONS AND ACTIONS

1. The ESSA concludes that the Program has a moderate environmental risk and moderate social risk. The systems are in line with the Core Principles and Key Planning Elements as defined in the World Bank Policy for PforR. The Program will require increased coordination at the national and state level on environmental and social aspects to further support their implementation. The process and criteria for monitoring, enforcement, and reporting on environmental and social measures will be part of overall Program reporting. The above requirements, processes, and systems will be included in the Program Operations Manual. Monitoring and supervision of the ESSA implementation will be a part of World Bank supervision

Program Exclusions

2. The following activities will be excluded from support under the proposed PforR Program expenditure based on environmental and social risks:

(i) Activities involving asbestos containing materials (AC roofing sheets, AC pipes, and so on) such as installation of GRPV, demolition, dismantling on AC roofs.

(ii) Any activity that may involve land acquisition or have potential involuntary resettlement will be excluded (screened out) from the Program boundary.

(iii) Use of child or bonded or forced labor or labor involved in any hazardous activities.

(iv) Destruction or damage to any physical and cultural resources or natural habitat or protected area.

Recommendations

3. The ESSA recommendations focus on strengthening the implementation arrangements, program procedures, risk mitigation practices/protocols and monitoring and reporting systems on E&S aspects, as well as core training programs to build the technical and operational capacity, and awareness generation. While most of the recommendations to be incorporated in the program operations manual (POM) and some are mainstreamed as part of result framework, a higher-level action is recommended as part of the program action plan (PAP).

4. The provisions of the existing environmental legal and regulatory framework are adequate but require enabling institutional and technical capacity to comply with and monitor, as well as generate awareness amongst consumers.

5. ***Solar Waste:*** It is essential to work closely with stakeholders, key being manufacturers, Ministry of Environment Forest and Climate Change (MoEFCC) and MNRE, at policy level to establish enabling environment to effectively implement the take-back policy of decommissioned panels. It should promote collection and storage of solar waste, and safe and efficient processing of stored waste. In the meantime, in it important to maintain information of installed solar capacity with details of panels, for accurate mapping of hotspots of solar waste generation.

6. A generic ***guidance checklist for addressing the Environment Health and Safety (EHS) requirements*** of the GRPV program can facilitate appraisal and periodic monitoring during installation and operation. An *additional guidance checklist for compliance* is also provided, to enable GRPV developers to understand the EHS requirements of the GRPV program and comply accordingly (Annexure 5).

7. ***Monitoring system*** of PMSG should be enhanced to monitor environmental management aspects related to GRPV. The system may be enhanced with use of above checklists and spatial mapping tools.

8. ***Social.*** As per the assessment against core principles 1, 3, 4 and 5, the national and state government have robust and enabling laws and policies to address the effects of the program and associated risks. The

challenges are largely in terms of ensuring the operationalization and enforcement of regulations. It is recommended to exclude certain activities which exacerbate risks and are outside the ambit of the Program. Section three provides a Screening Checklist to map and exclude high risk interventions under the proposed investments. The following actions are recommended:

a) Recruitment of at least one social specialist with comprehensive E&S risk management training, at MNRE.

b) Preparation of program specific Check Lists and Standard Operating Procedures; such as a) exclusion list, b) procedures for – (i) risk screening, categorization, guidance for avoiding and minimizing risks and impacts identified through screening, (ii) land management (which would, among others, include mitigation measures for impacts on non-titleholders, informal settlers and third party affected persons or groups, restrictions on use or access of land, land fragmentation), (iii) labor management, (iv) GBV and SEA/SH risk assessment, (v) assessing risks and impacts of program activities on tribal populations, (vi) stakeholder engagement (including, as and where required, activity specific GRMs), (c) templates for environment and social management plans (to be detailed for each activity based on activity specific risk classification), (d) Plans for building E&S risk management capacities of Empaneled Agencies and DISCOMs (e) templates for periodic reporting on management of program specific E&S risks and impacts

c) Updates/ Modifications to the Portal enabling disaggregated data on gender, vulnerability, socio-economic statues, etc.

d) Updates/ Modifications to Grievance Redressal Mechanism to include acknowledgement of complaints, resolution durations, escalation of complaints, feedback from complainants, etc. Increase the capacity of the GRM Portal to handle more complaints in view of the target 10 million consumers (presently about 150 tickets can be handled per day).

Program Action Plans

9. The assessment identified certain areas for improvement of the implementation of the environmental and social systems, which can be addressed through the following recommendations:

Table 8: Environmental and Social Program Action Plan

Action Description	Source	Responsibility	Timing	Completion Measurement
Screening of sites for exclusion checklist	ESSA	MNRE	Within 3 months of Program effectiveness	Screening criteria incorporated in PMSG's National Portal and Guidelines updated.
Adoption of EHS Guidelines	ESSA	MNRE	Within 6 months of Program effectiveness	Checklists incorporated in PMSG through updating relevant Guidelines and their dissemination through trainings and capacity building.
Development and adoption of policy/action plan to manage solar waste	ESSA	MNRE in collaboration with MoEFCC	Within 24 months of Program effectiveness	Policy/Action Plan prepared and disclosed on PMSG's website.

Annexure 1: Summary of the Current Status of Solar E-Waste in India and Next Steps for Environmental Management

1. The study *"Enabling a Circular Economy in India's Solar Industry: Assessing the Solar Waste Quantum"*, conducted by the Council on Energy, Environment and Water (CEEW) in collaboration with the Ministry of New and Renewable Energy (MNRE), provides a comprehensive analysis of India's solar waste landscape and its implications for sustainability. As India rapidly expands its solar capacity, the country faces an increasing challenge in managing end-of-life (EoL) solar panels and associated waste. The study highlights that India's cumulative solar waste could reach 19,000 kilotonnes (kt) by 2050, necessitating immediate policy intervention and infrastructure development. While the E-Waste Management Rules, 2022 include solar PV modules, the study underscores the lack of robust implementation frameworks, efficient recycling technologies, and regulatory enforcement. This calls for a systematic and circular economy-driven approach to ensure responsible waste handling, resource recovery, and environmental safety.

2. The study identifies key sources of solar waste, including EoL modules, damaged panels from installation and transportation, and repowering of existing systems. It estimates that 67% of India's solar waste by 2030 will originate from five states: Rajasthan, Gujarat, Karnataka, Andhra Pradesh, and Tamil Nadu. Despite these projections, India currently lacks commercial-scale solar recycling infrastructure, and most discarded panels end up in landfills or informal processing units, leading to potential environmental hazards. **(URL: https://www.ceew.in/publications/how-can-india-enable-circular-economy-with-solar-energy-waste-management-disposal)**

Current Status of Solar E-Waste in India

3. India's solar capacity has expanded significantly, reaching 73.32 GW as of December 2023. This rapid deployment is expected to generate substantial solar waste, with estimates indicating that waste could range from 0.1 to 0.7 million metric tonnes by 2030 and rise to approximately 19 million metric tonnes by 2050.

4. The major sources of solar waste include:

- End-of-life (EoL) solar modules (modules reaching the end of their operational lifespan).
- Damaged modules (from transportation, installation, or operational failures).
- Modules replaced during repowering (upgrading systems with newer technology).

5. Currently, India lacks a well-defined and efficient solar e-waste management infrastructure. Though the E-Waste Management Rules, 2022 include solar PV modules under the Extended Producer Responsibility (EPR) framework, clear guidelines on implementation, collection, and recycling are still required.

Challenges and Gaps in Solar Waste Management

- Lack of Recycling Infrastructure – India has no commercially operational solar panel recycling plants. Most solar modules end up in landfills or informal dismantling units.
- High Transportation Costs – Waste from large-scale solar projects in remote areas is expensive to transport to potential recycling units.
- Toxic Materials and Environmental Risks – Solar panels contain hazardous materials such as lead, cadmium, and tellurium, which can contaminate soil and water if improperly handled.
- Limited Awareness and Policy Enforcement – Many stakeholders are unaware of regulatory requirements, leading to poor compliance with e-waste rules.

- Economic Viability of Recycling – Current recycling technologies are expensive and limited to recovering low-value materials like glass and aluminum, while critical materials like silver and silicon are not efficiently recovered.

Next Steps and Recommendations

6. To address the growing challenge of solar e-waste, India must adopt a circular economy approach focused on reducing waste, enhancing recycling, and promoting sustainability.

- **Strengthen Policy Implementation** – The Ministry of Environment, Forest and Climate Change (MoEFCC) should issue detailed implementation guidelines for the E-Waste Management Rules, ensuring clear roles for producers, recyclers, and other stakeholders.
- **Develop Recycling Infrastructure** – The Ministry of New and Renewable Energy (MNRE) should support the establishment of solar PV recycling facilities and promote investment in advanced recycling technologies.
- **Encourage Producer Responsibility** – Solar module manufacturers should build collection centers and storage facilities for solar waste in compliance with EPR regulations.
- **Enhance Data Collection** – A centralized database of installed solar capacity should be maintained to track the lifecycle of solar panels and forecast waste generation more accurately.
- **Invest in R&D for Recycling Technologies** – Innovation in efficient and cost-effective recycling methods is crucial for recovering valuable materials like silicon and silver.
- **Promote Awareness and Capacity Building** – Government and industry players should conduct awareness campaigns for RWAs, DISCOMs, and other stakeholders on responsible solar waste management.

7. By taking these steps, India can develop a sustainable and efficient solar e-waste management ecosystem, ensuring long-term environmental and economic benefits.

Annexure 2: Estimated Solar PV Panel Waste from PM Surya Ghar: Muft Bijli Yojana

1. The PM Surya Ghar: Muft Bijli Yojana aims to install 40 GW of solar photovoltaic (PV) capacity across ten million households, assuming an average installation of 4 kW per household. This large-scale deployment of solar panels necessitates a strategic assessment of potential waste generation and recycling mechanisms to ensure sustainable disposal and resource recovery.

2. Assumptions for Estimation of Solar PV Panel Waste

- **Total Installed Capacity:** 40 GW (10 million households, each with 4 kW system)
- **Types of Panels:** Bifacial & Half-Cut Cell Panels (newer technologies)
- **Approximate Weight of PV Panels:** 25-30 kg per panel
- **Weight for a 4 kW System:** ~200-240 kg per household
- **Total Weight of Installed Panels:**
  - Lower estimate: 2.0 million metric tons (10,000,000 × 200 kg)
  - Upper estimate: 2.4 million metric tons (10,000,000 × 240 kg)
- **Additional Component Weight:** Mounting structures, inverters, wiring, and other accessories add further material that will require disposal and/or recycling.

3. Waste Generation Over Time

- **Early Replacements:** 10% of total installations may require replacement before 20 years (~0.2 million metric tons of waste).
- **End-of-Life Waste (20–25 years):** Total estimated PV panel waste is 2.2 million metric tons.

4. Composition of PV Panels & Recycling Potential PV panels are primarily composed of:

- **Glass (~75-80%)** – Highly recyclable
- **Aluminium (~8-10%)** – Easily recyclable
- **Silicon (Semiconductor Material ~3-5%)** – Can be recovered and repurposed
- **Copper, Silver, and Other Metals (~5%)** – Valuable for extraction and reuse

5. Given this composition, over 80% of panel materials can be successfully recovered if appropriate recycling mechanisms are in place.

6. **Regulatory Framework for Solar Panel Waste in India** the management and disposal of solar PV panel waste in India are governed by the **E-Waste (Management) Rules, 2022**, notified by the Ministry of Environment, Forest and Climate Change. Key provisions include:

- **Extended Producer Responsibility (EPR):**
  - Manufacturers and producers must register on the designated portal to ensure compliance and transparency.
  - Waste generated must be stored until 2034–2035 per CPCB guidelines.
  - Annual returns on e-waste management must be filed until 2034–2035.
- **Recycling Mandate:**
  - Recyclers must recover materials as specified by CPCB guidelines.

7. **Existing Recycling Infrastructure in India** India's solar panel recycling infrastructure is in a nascent stage, but some key initiatives include:

- **First Solar:** Operates a recycling facility in India capable of recovering over 90% of materials from processed modules.

- **Dynamic Industry Corp:** Plans to establish two recycling stations (north and south India) to tackle solar waste.
- **National PV Recycling Program:** Founded in 2016, this initiative comprises recycling and refurbishment providers approved by the Solar Energy Industries Association (SEIA).

8. **Challenges and Future Outlook** While these initiatives mark progress, challenges remain:

- **Limited recycling facilities:** Infrastructure must be scaled up to handle projected waste volumes.
- **High recycling costs:** Efficient material recovery processes need investment and technological advancements.
- **Regulatory enforcement:** Stronger implementation of e-waste rules is required for effective compliance.
- **Consumer Awareness:** Households and installers must be educated on end-of-life panel management and disposal practices.

9. To address these challenges, the government and industry stakeholders must collaborate to establish a robust framework for handling increasing solar panel waste. Investment in domestic recycling capabilities, extended producer responsibility enforcement, and incentives for recycling infrastructure will be crucial for sustainable solar energy expansion in India.

10. **Conclusion** The PM Surya Ghar: Muft Bijli Yojana is a transformative initiative for India's energy transition. However, with the massive deployment of solar PV systems, a strategic approach to waste management and recycling is essential. Proactive planning, regulatory adherence, and infrastructure development will ensure environmental sustainability while promoting a circular economy in the solar sector.

Annexure 3: Stakeholders Consultations

Individual User and RWA Society (Delhi NCR Region)

1. **Individual User:** The user has a load of 7KW and installed 3.24 KW solar rooftop PV by hiring the services of a registered vendor from the MNRE portal. He negotiated on the quote given by the vendor (M/S Solar Square). The installation was undertaken in Oct 2024 and took approximately 15 – 20 days to complete the process. The civil works and installation took approximately 2-3 days as per the user. generation guarantee to the end user as well. The subsidy amount was also received by the user in his account within 30 days of the installation. (INR 30000 per kW up to 2 KW than 18000 per KW up to a maximum of INR 78,000) + (INR 6000 from Delhi Government at the rate of 2000 per KW), + generation-based incentive from Delhi government.

2. The agreement signed between the user and vendor also provides for five-year warranty and maintenance services wherein, the maintenance team visits the site at least once/twice a month for the cleaning and if required by the user, the frequency can be increased as well. The vendor has also provided basic awareness on fire safety and associated hazards of the system. A fire extinguisher was also installed on site by the vendor. There is no insurance cover for the installed systems in place either by the user himself or by the vendor.

3. Interactions with the user indicated that certain facilities, such as the provision of fire extinguishers and regular cleaning services, are implemented by the vendor as best practices but are not mandated under the template contract agreement. Furthermore, the user confirmed that all interactions with DISCOM staff and procurement-related activities were carried out solely by the vendor, including due diligence and safety inspections of the rooftop site. However, no official records of the due diligence conducted by the vendor's staff have been maintained, nor has any report been submitted to MNRE. This lack of documentation results in a significant gap in the systematic and standardized approach to rooftop assessments, leaving the process largely at the discretion of the vendor's staff.

4. **RWA Society:** The installed solar rooftop system has a DC capacity of 60 kW and an AC capacity of 50 kW, with solar generation recorded monthly by BRPL. The installation, undertaken by Vidhi Power Ventures, was completed within two months after obtaining necessary clearances from BRPL. The vendor conducted all due diligence, including obtaining structural design approval from the RWA and conducting a preliminary physical assessment of the rooftop. The system comprises bifacial 575W Topcon panels.

5. In this case, the structural designs proposed by the vendor were reviewed and approved by the RWA to ensure the safety of the installation. Additionally, the RWA secured residents' consent through a General Body Meeting (GBM) before proceeding with the registration and installation of the solar PV system under the government scheme. No additional administrative clearances were required.

6. The society, consisting of six towers, has installed the initial 60 kW solar system on the rooftops of three towers and plans to expand the capacity by another 60 kW across the remaining towers, subject to the project's success.

7. Cleaning is a critical aspect of maintenance, conducted three times a month to ensure optimal system performance. Each cleaning cycle consumes approximately 3,000–4,000 liters of water. While no cleaning agents are regularly used, exceptions are made in cases of bird droppings. Lack of regular cleaning can lead to a 10–20% drop in efficiency, negatively impacting the system's commercial viability and extending the payback period.

8. Currently, there is no dedicated water pipeline for cleaning; instead, water from the fire tank hose is utilized. Establishing a dedicated pipeline could improve efficiency and reduce dependency on fire safety resources.

9. Despite six months having passed since installation, the RWA has yet to receive the ₹11 lakh subsidy from MNRE. Timely disbursement of such subsidies is critical to maintaining confidence among RWAs and encouraging further participation in government solar programs. The total system cost was ₹29 lakh, and no loans were availed for the project.

10. Similar to individual user installations, the vendor in this case has proactively provided structural designs for validation, addressed safety concerns, installed fire extinguishers, and facilitated regular cleaning services as good practices for the RWA society. However, these practices remain vendor-specific and are not mandated under any standard framework.

11. To ensure consistency and reliability across all solar rooftop installations, it is recommended that such best practices be standardized through the adoption of **Standard Operating Procedures (SOPs)**. This will help streamline installation processes, improve safety and maintenance standards, and enhance the overall efficiency of rooftop solar projects.

12. **Stakeholder Consultations – Vendors (operating in more than 3-4 states):** Stakeholder consultations were undertaken on a sample basis with the RWA society members, individual residential owners and vendors during the field visit. In addition to these, a hybrid mode (Virtual & face to face participation) stakeholder consultation was undertaken with the vendors with the objective was to seek feedback on the constraints or challenges faced by developers and aggregators in their work, and to learn about their approach to management of environment and social issues. Findings from the interactions are summarized below in subsequent sections.

13. The key participants, comprising representatives from 12 solar system installation firms/vendors, included PAN India vendors currently implementing the government program across multiple states. Most of these firms operate in at least four states and are registered as PAN India vendors. They primarily work through channel partners, including those from outside the solar sector who are interested in supporting the program despite lacking prior experience. Their participation is driven by the sector's growth potential.

14. Some firms have also obtained ISO and CII certifications as part of their best practices, even though these certifications are not mandatory under the MNRE program. However, these good practices are not uniformly adopted across all vendors and vary significantly.

15. Firms like Tata Power leverage their own app to validate roof structures and ensure safety using AI technology. The system provides an instant sanction or rejection based on quality standards, requiring any negative markings to be rectified until a 100% score is achieved. A warranty certificate is issued only upon meeting these standards, making it an integral part of the incentive structure for channel partners. As a result of these initiatives, they have received minimal complaints in the residential sector. O&M activities are being carried out separate teams

16. The key environmental safeguard compliance gaps at each stage of the government's v program implementation include:

Planning Stage:

- Site Selection: Lack of clear guidelines on environmental considerations such as roof strength, shadow-free area, and ecological impact.

- Standardization: Absence of uniform environmental standards across states, leading to inconsistencies in implementation.

Procurement and Supply Chain:

- Supply Chain Disruptions: Delays and price fluctuations in the supply chain may result in the use of substandard or non-compliant materials. The high demand for domestically manufactured solar PV panels, driven by multiple government subsidy programs, has led to supply shortages and rising market prices, making affordability a challenge for end users.

Installation Stage:

- Worker Safety: Gaps in enforcing mandatory use of personal protective equipment (PPE) and adherence to safety protocols.
- Roofing Materials and Drainage: Inadequate assessment of roofing material compatibility and proper drainage systems, leading to potential structural issues. Some of the vendors adopt a SOP for their teams for installation but it's not a mandatory practice adopted by all vendors.
- Access to the rooftop is assured either with a no-objection certificate (NoC) from the rooftop owner, or a lease agreement giving the developer rights to the roof. Water availability for panels is another consideration, as approximately 0.5 litres of water per panel per month is required. So, making water available on the roof is also part of the PPA
- Waste Management: Lack of guidelines for disposal and recycling of installation-related waste (e.g., packaging materials, damaged panels). The vendor takes responsibility in some of the cases.

Operations and Maintenance:

- Fire and Electrical Safety: Absence of stringent fire safety measures such as fire-resistant materials, automatic shut-off systems, and training for homeowners.
- Insurance and Risk Management: Limited availability of comprehensive insurance covering fire, electrical hazards, and natural disasters.
- Maintenance and Monitoring: No clear framework for periodic environmental inspections to ensure ongoing compliance with safety and performance standards.

Decommissioning and End-of-Life Management:

- Solar Panel Disposal: No clear policy on the responsible disposal or recycling of end-of-life solar panels, batteries, and inverters.
- E-Waste Regulations: Potential non-compliance with India's e-waste management rules, leading to environmental hazards.

Other Compliance Gaps:

- Lack of Environmental Training: No structured program for training workers, vendors, and DISCOM officials on environmental safeguards.
- Public Awareness Deficiency: Insufficient efforts to educate end users on the environmental benefits and risks associated with solar rooftop installations.

17. Certain large vendors, such as TATA, Luminous, and Redington, have established Standard Operating Procedures (SOPs) that incorporate fundamental environmental safeguards, including worker safety, responsible waste management, and adherence to site selection guidelines. These practices help mitigate potential environmental and occupational hazards during solar rooftop installations.

18. The solar roftop installation in the RWA society demonstrates a proactive approach by the vendor in ensuring structural safety, securing necessary approvals, and incorporating good practices such as fire

safety measures and regular maintenance. The successful execution of the project within a short timeframe, along with the society's plans for expansion, highlights the growing acceptance of solar energy solutions under government schemes.

19. However, several critical gaps have been identified in the implementation process. The absence of standardized maintenance protocols, such as dedicated water supply for cleaning, affects operational efficiency and system longevity. Additionally, delays in subsidy disbursement, as observed in this case, can discourage broader participation in solar adoption. Ensuring timely financial incentives is crucial to maintaining stakeholder confidence.

20. A key concern is the lack of uniformity in environmental and safety compliance across vendors. Stakeholder interactions revealed that while some vendors implement best practices, smaller vendors and local installers often lack standardized protocols. This inconsistency leads to variations in safety adherence, environmental risk management, and overall quality assurance. The absence of a structured and regulated framework results in gaps in environmental safeguard compliance, which could have long-term implications for sustainable solar deployment.

21. To address these challenges, it is essential to establish and enforce **industry-wide Standard Operating Procedures (SOPs)** that mandate uniform environmental and safety standards. This will help ensure responsible and sustainable implementation of solar projects, mitigate operational risks, and enhance overall project reliability.

22. Another key challenge highlighted by vendors, though not directly related to environmental compliance, is the high overall cost of solar systems and the return on investment for end users. Non-DCR panels are priced at approximately INR 12.50 per kW, while DCR panels cost nearly double at INR 25.2 per kW due to supply-demand imbalances in India. Additionally, multiple government programs offer subsidies on DCR panels, but supply remains limited. The subsidy is only beneficial for installations below 3-4 kW per user, whereas for systems above 7 kW, it becomes economically unviable to avail of the subsidy.

23. Social. The following was the initial feedback from the participants on the Solar Program:

- The Public Sector Banks are very conservative in giving loans to even for small projects (both retail/ domestic and distributed solar projects), with delayed approvals and final payments. The banks say they are initiating measures to improve their performance. Whereas the Non-Banking Financial Corporations are doing the loan processing in about seven days and servicing the customers as per the requirments. When it comes to the public sector banks, at least two or three visits are required to the banks for loan approval and for payments; many customers' final 30% of the payments are delayed.
- Improving awareness of the participating banks about the program and need for swift support to customers is important.
- All the banks shoud have same guidelines and requirements for issue of solar loans.
- Presently there are around 12,000 solar channel partners; out of these 90% are new entrant with potential for becoming partners.
- These partners lack bandwidth to reach out to customers and to create awareness about the solar poject.
- For the empanelled vendors one or two kilowatts projects are not attractive, but above 5 kilowatts would interest them more; as one or two kilowatt projects will amount to only about INR10,000 of profit for the vendors.
- These days the panel rates have been sky racketing suddenly. Normally the non-DCR panels can be procured at about INR 6/Wp, whereas the DCR panels are sold at about INR 25/Wp. It appears an artificial demand is created and customers are not getting panels in the open market as the demand out strips supply.

- The vendors inform that the DCR manufacturers are exporting the panels and these exports need to be banned till the project goals and domestic market needs are met.
- There are several complaints about the solar portal; some of the key complaints are a) the portal is either permaneltly or regularly down, b) payments are over delayed, c) the menu to download customer data are disabled, d) there are no replies from MNRE for the complaints, e) no feedback is collected from customers, f) registrations take a lot of time, g) for filing any grivances the vendors and customers are unable to reach MNRE, etc,. Since some customers have not received their subsidies, they are blaming the vendors for this and sending legal notices to the vendors and also making complaints to the Chief Ministers, Prime Minister, MLAs, MPs, etc. and so many other people.
- Obtaining the required certificates after new guidelines are issued is posing a problem for the customers and also for the vendors. Customer Support is required at MNRE level. Presently no replies are given to any complaints from MNRE.
- Only UP, Maharashtra Gujarat, etc. states are doing very well in achieving targets, whereas the East and North East states are not performing well. This need to be taken care of by providing awareness to the beneficiaries and removing all the obstacles blocking the progress.
- Presently, the project is around 80% in urban areas and around 20% in rural areas; this need to be changed and more rural projects need to be taken up to meet the target.
- The Vendors say for the safety of the workers, they provide them with PPE kits, all the required tools, instruments for measuring us voltage and resistance, etc. are provided.
- The vendors say that most of the workers are either ITI certificate holders or diploma holders or had some training in the Solar installation. The workers are also given some induction training. Some of the vendors are CII certified.
- The Vendors say they have registraion. They say there are some small scale vendors who who may not have such kind of registrations are certifications presently.
- Some of the DISCOM's don't even inspect this sites for monitoring/ inspection/ commissioning/ certification.
- The Vendors say that the empanelled vendors are having licenses as per the state level norms. The insurance is something that some vendors have and some vendors or yet to take depending upon the kind of contracts they have with their workers.
- Registration of migrant labor is not taken up by the vendors.
- The vendors say they are paying the workers for accommodation during outstation duties when they go on site.
- Safety tests such as Vetigo Test for workers working at heights are yet to be introduced. As per the vendors, in general there is some training or induction training given to the workers. They say so far no accidents are reported.
- The vendors say no female workers are working on site, but they work as office staff at the offices. They say GBV training is given by some of the vendors to their workers. But they do not have any GBV training material of their own. They have to start formation of ICs.
- The vendors inform that an informal internal GRM is in place for workers.
- The vendors say they will be sharing the checklist that DISCOMs use for approving the project.
- Some of the vendors say they have manuals and standard operating procedures for health and safety, but some of them do not have. They will share these with the bank.

Annexure 4: Sub-Project Due Diligence/Appraisal

Stage Wise Due Diligence	
Stage	***Environmental and Social Safeguards Considered***
Initial Stage / Residential User Registration stage	The residential user registers on the web portal by sharing his electricity consumer number, name, and mobile number. The complete process is available online in the consumer user manual. (https://pmsuryaghar.gov.in/assets/files/Consumer%20User%20Manual%20PMSG%20MBY%20Portal%202.0%20Final.pdf) The consumer details captured during the process include his/her name, mobile number for OTP verification and electricity consumer number.
Vendor shortlisting and selection by the Residential User	Post acceptance of the registration, the residential user shortlists the vendor. Vendor undertakes technical and contractual feasibility covering: -. Rooftop ownership details (with a copy of ownership documents) of the rooftop where project is proposed to be installed. Access rights to the rooftop for installation & O&M stage is validated by the vendor. Rooftop profiling and site assessment (including structural stability) to analyze aggregate rooftop area, rooftop area suitable for PV project development, potential for PV capacity is assessed by the vendor. All necessary clearances and approvals (fire safety measures, earthing) are required to be obtained before installation. Preliminary go ahead from the local DISCOM based on the position of power evacuation infrastructure in the area. Vendor undertakes such permissions on behalf of the end user. Vendor signs an agreement with end user for installation, and O&M for at least five years as per the standard agreement approved by MNRE. *Gaps in the due diligence include:* *The "Rooftop Solar PV (Installation & Maintenance) Training Manual" incorporates Occupational Health and Safety (OHS) aspects primarily through guidelines on safe installation, maintenance, and troubleshooting of solar PV systems. It covers essential safety measures such as the use of personal protective equipment (PPE), fall protection for rooftop work, and fire safety precautions related to electrical components. Additionally, it includes protocols for handling batteries, inspecting system components, and ensuring proper grounding to prevent hazards. However, the manual focuses mainly on workplace safety rather than a broader Environmental, Health, and Safety (EHS) framework.* *No specific details on the environmental performance, rating by previous end users, certification details (ISO certifications, certified grade-1 electrical contractor, labour licence, OHS practices etc) are included in the portal for the vendors.*

	Aspects related to availability of water, rooftop drainage aspects, activity related to pruning of trees (if required) are not specifically covered in the guidelines for installation of solar pv systems. *There are no specific checks to monitor/verify whether the end-user property violates any bylaws, is situated in environmentally sensitive areas, or is on encroached land. Availability of an existing electrical connection is the only requirement to avail benefits under the program.*
Financial & Subsidy	Within the main portal, a direct link is provided to the Jan Samarth Portal, a government-backed credit facilitation platform designed to offer seamless access to various loan schemes. Through this integration, users can explore available loan options, compare interest rates from multiple financial institutions, and apply for credit online without needing to visit bank branches physically.
Inspection by the DISCOM	Post installation of the solar pv systems, the DISCOM undertakes physical inspection of the systems to ensure electrical safety before granting permissions for commissioning. *Gaps in the due diligence include:* *No technical due diligence is performed for systems up to 10 kW as per the guidelines. Discom's only verify the availability of power evacuation infrastructure and ensure grid safety, while the responsibility for system installation lies solely with the vendor. However, there is no reporting mechanism to validate compliance.*
Commissioning of the systems	Discom's conduct physical inspections to ensure compliance with quality and safety norms. Upon successful installation and inspection, the system is commissioned with a net meter connection.
Post approval process	The agreement specifies the duration (minimum of 5 years) and terms of warranties for various components, such as solar panels and inverters. Vendors are also responsible for providing maintenance services during the warranty period to ensure optimal system performance. MNRE/REC disburse the subsidy directly to consumers. *Gaps in the due diligence include:* *The program does not specify details regarding the maintenance and upkeep of rooftop solar systems beyond the initial five-year Operations & Maintenance (O&M) period. There is no structured mechanism for monitoring system performance, ensuring vendor accountability, or providing support for repairs and replacements after the mandatory O&M duration. This gap raises concerns about the long-term sustainability and efficiency of installed systems.*
Vendor Training	To address the gaps in the skilled manpower and safety aspects, the Ministry of Skill development and entrepreneurship under the skill India program has prepared a "Rooftop Solar PV (installation and Maintenance)". The document is a training

	manual for rooftop solar PV installation and maintenance, developed under the NSQF Level 3 framework by the Ministry of Skill Development and Entrepreneurship, India. URL: https://pmsuryaghar.gov.in/assets/files/Vendor%20User%20Manual%20PMSG%20MBY%20Portal%202.0%20Final.pdf It provides theoretical and practical modules covering solar energy fundamentals, photovoltaic systems, installation procedures, maintenance, safety, and testing standards, aiming to skill technicians for the PM Surya Ghar initiative. *Gaps in the due diligence include:* *The training manual does not cover environmental aspects of emergency preparedness and response planning, incident reporting and protocols, approvals and necessary clearances, onsite storage disposal of hazardous materials, first aid provisions*

Annexure 5: Guidance Checklist and Consumer Education for verification of adequacy on Environmental, Health, and Safety (EHS) requirements during appraisal and monitoring (Installation and Operation phases)

The checklist includes three parts – Go/no-go criteria, for residnetial consumers/project developers, and Safety guidelines to be adhered to by vendor/RESCO/ULA

1. Go / No-Go Criteria

S. No.	EHS Requirement of GRPV Program	Yes/ No
1.	Confirm that Roofing material where the rooftop solar system is installed does not contain any carcinogenic material like broken or dilapidated Asbestos.	

2. For Residential Consumers / Project Developers

S. No.	EHS Requirements of GRPV Program	Status (State Yes/No/Not Applicable)	Guidance for ensuring compliance of EHS requirements by MNRE	Review and Monitoring by MNRE for adequacy and compliance of EHS requirements
Proposal Appraisal Phase				
1.	Whether the GRPV proposal requires lopping/pruning of tree branches to ensure a shadow-free area on the roof. If yes, state whether permissions are obtained from competent authorities for periodic lopping/pruning of trees		If yes, check validity and conditions imposed on proponent by a competent authority, if any. If not, ensure first disbursement is released subject to the submission of valid permissions for loping /Pruning of trees.	Review compliance with permissions including conditions, if any by proponent through site inspections by SIA or MNRE's staff.
2.	Whether the GRPV proposal is on building that is in protected area including cultural heritage, demarcated forest area, reserve area, etc.		If yes, check conditions imposed on proponent by concerned line department/competent authority and get necessary clearances.	Review compliance with permissions including conditions, if any, by proponent through site inspections by SIA or MNRE's staff.

S. No.	EHS Requirements of GRPV Program	Status (State Yes/No/Not Applicable)	Guidance for ensuring compliance of EHS requirements by MNRE	Review and Monitoring by MNRE for adequacy and compliance of EHS requirements
3.	Whether roof rights have been secured		If yes, please verify the lease agreement/draft lease agreement/title deed for establishing clear rights over the roof for installation and operations.	Review compliance to permission including conditions if any by proponent through Legal Counsel or SIA or MNRE's staff..
4.	Whether access is available on a 24 X 365 (all days of the year irrespective of public holidays and Sundays). Whether permissions from the owner (in case of tenant) is available to access the roof through the existing staircase. whether external access will be required.		If not, seek details of alternative safe access along with permission from owner. Seek details of arrangements made for safe lifting of the materials to rooftop through existing staircase or temporary/ permanent external access.	Review the safety of the alternate access to the roof through site inspections by SIA or MNRE's staff.
5.	Whether structural safety of the building, present condition of roof for leakages and/ or cracks and adequacy of roof drainage has been assessed		Seek a structural safety and roof condition certificate from a certified/approved Chartered Engineer / Architect/ Competent person along with an action plan for rectifications and responsibilities if any required. If not, ensure the certificate is submitted by proponent prior to the first disbursement of the loan.	Check the validity, review the adequacy of arrangements through by SIA or MNRE's staff.
6.	Whether the consent[19] from residents / owners/ general body has been secured? Whether the residents are informed about the timelines of the construction process?		If yes, please verify the consent document arrangements for maintaining secure and non-intrusive access to the installation site made and agreed with roof user/RWA. If not, ensure the first disbursement is released	Review compliance with permissions by project proponent through site inspection by SIA or MNRE's staff.

1. _____

[19] Include an arrangement for maintaining secure and non-intrusive access to the installation site made and agreed with roof user/RWA.

S. No.	EHS Requirements of GRPV Program	Status (State Yes/No/Not Applicable)	Guidance for ensuring compliance of EHS requirements by MNRE	Review and Monitoring by MNRE for adequacy and compliance of EHS requirements
			subject to the submission of a valid document. The developer should also submit the timelines for construction activities	
7.	Whether proposal includes estimated water requirements for washing of panels and dependable arrangements to draw or share water from the same water connection or overhead tanks with owner of the building		Seek details of water requirements and its sources along with required permissions from competent authorities, if any required.	Review the adequacy of arrangements through monitoring by SIA or MNRE's staff.
8.	Does the loan include financial assistance for batteries?		If yes, ensure that undertaking is available for compliance with current Batteries (Management) Rules 2021 and amendments thereof	Check whether an agreement with authorized recycler is in place.
Installation and Operation Phase				
9.	State whether the supplier / installer / RESCO developer has made / agreed for any arrangement with manufacturer to take back damaged /Discarded panels, Batteries, etc.		Seek details of take-back arrangement with the manufacturer, and in case such arrangement is not there with manufacturer stipulate condition in the sanction that the disposal of the panel should be as per applicable local law for discarding such hazardous waste.	The undertaking will be taken from the proponent for compliance of the condition.
10.	Electrical Safety Approval: Whether earthing of all plant and equipment / components under GRPV as per Indian Electricity Act,1956 and amended up to 2000 has been made and tested by an approved competent agency		Seek certification from Chief Electrical Inspector to Government (CIG), if applicable for the rooftop solar system size as defined in the respective state (or central) regulations.	The undertaking will be taken from the proponent for compliance of the condition.
11.	Whether the end consumers have been sensitized to the potential safety issues in installing solar rooftop plants		Share material about potential safety hazards and likely restriction to activity in the areas where plant and machinery are to be installed/stored.	Review compliance of safety measures through site inspection by LIE or SBI's staff.

3. Safety guidelines to be adhered to by vendor/RESCO/ULA

S. No.	EHS Requirements of GRPV Program	Status (State Yes/No/Not Applicable)	Guidance for ensuring compliance of EHS requirements by MNRE	Review and Monitoring by MNRE for adequacy and compliance of EHS requirements
1.	Whether the proponent / installer has accreditation of ISO 14000, OHSAS 18001 or has received any recognitions for environmentally friendly initiatives or best EHS practices		If yes, seek details of valid certifications and or recognitions. Accreditation(s) give an indication to institutional the capacity of the proponent to EHS requirements.	
2.	All safety provisions like provision of rubber mats, electric shock chart, first aid box, fire extinguishers to handle all types of fire (ABC type of required capacity), sand buckets, etc. should be provided/installed at appropriate locations		Seek details of safety measures/provisions mandatorily provided prior to testing, trial run and commercial operations of GRPV facility.	Assess adequacy and review the safety provisions including exit routes provided and procedures followed during site inspections and monitoring by SIA or MNRE's staff.
3.	The provisions to provide safety wear like boots, hard hats (helmets), gloves, safety belts for personnel while working at heights among others have been included in the proposal.		Seek details of safety measures/provisions mandatorily provided to all workforce deployed on-site to ensure the safety of personnel at work.	Assess adequacy and review the safety provisions provided and procedures followed during site inspections and periodic monitoring by SIA or MNRE's staff.
4.	All personnel deployed for Installation / Operation and Maintenance are provided with basic training in first aid and firefighting.		An undertaking from the proponent that they will ensure that personnel deployed for Installation / O&M has a basic knowledge about first aid and fire-fighting instruments	
5.	All personnel deployed for Installation / Operation and Maintenance (Unskilled, semi-skilled		An undertaking from the Proponent, that they will ensure compliance of applicable Minimum Wages Act.	

S. No.	EHS Requirements of GRPV Program	Status (State Yes/No/Not Applicable)	Guidance for ensuring compliance of EHS requirements by MNRE	Review and Monitoring by MNRE for adequacy and compliance of EHS requirements
	and skilled) should be paid at minimum wages as per applicable Minimum Wages Act of Government of India.			
6.	All personnel deployed for Installation / O&M are covered under workmen compensation insurance policy, EPF (Employee Provident Fund) Act, Gratuity Act etc. as may be applicable or relevant		An undertaking from proponent they will ensure that all personnel deployed for Installation/ O&M personnel will be covered with workmen compensation insurance policy and are provided with benefits of any other applicable acts.	The adequacy of insurances to be checked by SIA or MNRE's staff.
7.	End consumers have been sensitized to the potential safety issues in installing solar rooftop plants		Share material about potential safety hazards, and likely restriction to activity in the areas where plant and machinery are to be installed/stored.	Review compliance of Safety measures through site inspection by SIA or MNRE's staff.